Differential Equations and the Oscillator

One equation, solved completely — and then found hiding inside everything else in the book.

WHERE WE ARE

Every law of physics in this book is a differential equation. Newton's second law, Maxwell's equations, the geodesic equation, the Schrödinger equation, the Klein–Gordon and Dirac and Yang–Mills equations — all of them relate a quantity to its own rate of change. That is not a stylistic preference. It is what a *law* is: a statement that the way something is now determines the way it will be an instant from now. Physics is the study of local rules; a differential equation is a local rule written down.

So solving them is most of what physicists do, and this chapter solves the one that matters most. It is embarrassingly simple —

$$\ddot{x} = -\omega^2 x$$

— and Chapter 0.3 already told you why it is unavoidable: expand any potential about a stable equilibrium and the first surviving term is quadratic, so **every stable system, looked at closely enough, is a harmonic oscillator**. That result is the reason this chapter is long. We are not studying a toy. We are studying the universal local model of stability, and the last section builds the ladder from *two masses on springs* to *a field is infinitely many oscillators* — which is the single most important structural bridge in this book, the one Chapter 5.3 walks across when it quantises a field.

Along the way one structural fact will be stated louder than anything else here: a linear differential equation is an **eigenvalue problem**, and the exponential is the eigenfunction of $\mathrm{d}/\mathrm{d}t$. Solving a constant-coefficient linear ODE *is* diagonalising the derivative. Chapters 0.9, 4.5 and 5.3 are that sentence, three times, in different costumes.

TOOLS YOU'LL NEED — Chapter 0.1: the derivative, the chain rule, and e defined by $\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{e}^t = \mathrm{e}^t$; the licence to separate differentials. Chapter 0.3: Taylor expansion, the small-oscillations result, and Euler's formula $\mathrm{e}^{\mathrm{i}\theta} = \cos\theta + \mathrm{i}\sin\theta$ — used constantly. Chapter 0.4: vector spaces, basis, dimension, linear maps. Chapter 0.5: eigenvalues and eigenvectors, the spectral theorem for symmetric matrices, and the warning about *defective* matrices — §3 and §7 each cash one of these. Chapter 0.6: the Hessian and its eigenvalues as principal curvatures.

1 · What a differential equation is

An **ordinary differential equation** (ODE) is a relation among an unknown function $y(t)$, its derivatives, and t itself. "Ordinary" means one independent variable; when there are several — a field depending on x, y, z, t — you get a *partial* differential equation, which §7 will produce for you at the end.

Three words classify almost everything we will do.

Order is the highest derivative appearing. Newton's law is second order, because force determines acceleration; this one fact is why you always get *two* constants of integration in mechanics, and why "where is it and how fast is it going" is the right amount of information to specify a trajectory.

Linear means the unknown and its derivatives appear only to the first power, never multiplied together, never inside a sin or a square root. The general linear ODE of order n is

$$L[y] \;\equiv\; y^{(n)} + a_{n-1}(t)\, y^{(n-1)} + \cdots + a_1(t)\, \dot{y} + a_0(t)\, y \;=\; f(t). \tag{0.8.1}$$

The coefficients $a_i(t)$ may do whatever they like; what matters is that y enters linearly. The point of the notation is that L is a **linear map** in the sense of Chapter 0.4: $L[\alpha y_1 + \beta y_2] = \alpha L[y_1] + \beta L[y_2]$, because differentiation is linear and so is multiplication by a fixed function. Everything in §3 follows from that single observation.

Homogeneous means $f \equiv 0$ — no source term, nothing driving the system from outside. An inhomogeneous equation has a source, and §2 and §6 are about what a source does.

Initial conditions, and the mathematical content of determinism

An ODE by itself has infinitely many solutions. To pick one you supply **initial conditions**: the value of y and of its first $n-1$ derivatives at one instant t_0. That count is not a convention — it is exactly what the following theorem says is needed, and exactly what it says is enough.

⚑ QUOTED, NOT PROVED — PICARD–LINDELÖF

This is the one result in this chapter we take on faith. *Let $F(t, \mathbf{y})$ be continuous in t and **Lipschitz** in $\mathbf{y}$ near a point $(t_0, \mathbf{y}_0)$ — meaning there is a constant K with $\|F(t, \mathbf{u}) - F(t, \mathbf{v})\| \leq K\,\|\mathbf{u} - \mathbf{v}\|$ for all $\mathbf{u}, \mathbf{v}$ in some neighbourhood. Then the initial-value problem*

$$\dot{\mathbf{y}} = F(t, \mathbf{y}), \qquad \mathbf{y}(t_0) = \mathbf{y}_0$$

has exactly one solution on some open interval containing t_0. The proof constructs the solution as the limit of the iterates $\mathbf{y}_{k+1}(t) = \mathbf{y}_0 + \int_{t_0}^{t} F(s, \mathbf{y}_k(s))\,\mathrm{d}s$ and shows the Lipschitz condition makes that map a contraction. It is a fixed-point argument, it is genuinely elementary, and it is in every analysis text; we are quoting it because building it here would cost a chapter and buy no physics.

Why it covers everything. The theorem is stated for a first-order *system*, but that is no restriction: given an n-th order equation, set $\mathbf{y} = (y, \dot{y}, \ddot{y}, \ldots, y^{(n-1)})$ and the equation becomes a first-order system for $\mathbf{y}$, whose last component's derivative is supplied by the original equation. So n initial numbers is precisely one initial vector, and the theorem applies verbatim.

The hypothesis has teeth. Drop Lipschitz and uniqueness dies. The equation $\dot{y} = 3y^{2/3}$ with $y(0) = 0$ is solved by $y \equiv 0$ and also by $y = t^3$ — and also by anything that sits at zero for a while and then takes off. The right-hand side is continuous, but its slope blows up at $y = 0$, so no K exists. Physical laws are built to be Lipschitz; when a model of yours has a non-unique solution, look for the missing physics.

Now read the theorem as a physical statement rather than an analytic one. It says: **the state now determines the state later, uniquely.** That is determinism, and it is not a philosophical posture bolted onto physics — it is a theorem about the equations, given that the equations are of this form. The "state" of a classical system is exactly the list of numbers the theorem requires, which for mechanics is position *and* velocity, and that list is what Chapter 1.3 will call a point of **phase space**.

Two places where this gets interesting, both flagged now and paid later. In §8 we meet systems that are perfectly deterministic and still unpredictable: the solution exists and is unique, but neighbouring initial conditions separate exponentially, so a finite-precision measurement of the present buys you only a finite time of usable future. That is *chaos*, and it is a statement about sensitivity, not about determinism. And in quantum mechanics the determinism survives intact in a place people rarely look: the Schrödinger equation $\mathrm{i}\hbar\,\partial_t\,|\psi\rangle = \hat{H}\,|\psi\rangle$ is a first-order linear ODE for the state, so the state evolves as deterministically as a planet. The randomness in quantum mechanics enters only at measurement, and Chapter 4.8 is about exactly where that seam is.

2 · First-order linear equations

Start where the mathematics is easiest and the physics is already familiar.

2.1 · Separation of variables, done honestly

If the equation has the form $\dot{y} = g(y)\,h(t)$ — the right-hand side factorises into a piece depending only on y and a piece depending only on t — then it can be integrated directly. The usual manipulation is to write $\mathrm{d}y/g(y) = h(t)\,\mathrm{d}t$ and integrate both sides, which is the "deliberate abuse" that Chapter 0.1 warned you was legitimate. Here is why it is legitimate, in one line. Let G be an antiderivative of $1/g$. Then by the chain rule,

$$\frac{\mathrm{d}}{\mathrm{d}t}G\big(y(t)\big) \;=\; G'(y)\,\dot{y} \;=\; \frac{\dot{y}}{g(y)} \;=\; h(t), \tag{0.8.2}$$

so $G(y(t)) = \int h(t)\,\mathrm{d}t + C$. No differentials were harmed. Separation of variables is the chain rule read from right to left.

The example you already know: $\dot{y} = -ky$ has $g(y) = y, h(t) = -k$, so $G(y) = \ln|y|$ and $\ln|y| = -kt + C$, giving $y = y_0\mathrm{e}^{-kt}$. Chapter 0.1 got this by recognising the defining property of e; we now have it as a special case of a method.

2.2 · The integrating factor, derived

Separation fails the moment there is a source: $\dot{y} = -ky + f(t)$ does not factorise. The trick that works is one of the most quietly clever manoeuvres in elementary mathematics, and it is worth deriving rather than remembering.

Take the general first-order linear equation

$$\dot{y} + p(t)\,y \;=\; f(t). \tag{0.8.3}$$

We would like the left-hand side to be a single derivative, because then we could integrate it. It is not — but suppose we multiply the whole equation by some function $\mu(t)$ and *ask* what μ would have to be for the result to be exactly $\frac{\mathrm{d}}{\mathrm{d}t}(\mu y)$. Expanding that target with the product rule,

$$\frac{\mathrm{d}}{\mathrm{d}t}(\mu y) \;=\; \mu\,\dot{y} + \dot{\mu}\,y \qquad \text{versus} \qquad \mu\dot{y} + \mu p\,y. \tag{0.8.4}$$

The two agree for every y precisely when $\dot{\mu} = p\,\mu$ — which is the separable equation we just solved. So μ is forced:

$$\mu(t) \;=\; \exp\!\left(\int^t p(s)\,\mathrm{d}s\right). \tag{0.8.5}$$

Nothing was guessed. We asked for the function that makes the left side a total derivative and the demand determined it. Multiplying (0.8.3) by μ and integrating from t_0 to t now gives the general solution outright:

$$y(t) \;=\; \frac{1}{\mu(t)}\left[\mu(t_0)y(t_0) + \int_{t_0}^{t} \mu(s)f(s)\,\mathrm{d}s\right]. \tag{0.8.6}$$

2.3 · Exponential decay with a source — and a Green's function in embryo

Specialise to constant $p = k$, so $\mu = \mathrm{e}^{kt}$, and set $t_0 = 0$:

$$\boxed{y(t) \;=\; \underbrace{y(0)\,\mathrm{e}^{-kt}}_{\text{memory of the start}} \;+\; \underbrace{\int_0^t \mathrm{e}^{-k(t-s)} f(s)\,\mathrm{d}s}_{\text{memory of the source}}} \tag{0.8.7}$$

Stare at the integral. It says: the source deposited an amount $f(s)\,\mathrm{d}s$ into the system at time s, that deposit has been decaying ever since for a duration $t - s$, and the present value is the sum over all past deposits of what is left of each. The system is a *memory* with a fading kernel, and the kernel is the response to a single sharp kick.

This is called the **Duhamel** form, and the function

$$G(t,s) \;=\; \mathrm{e}^{-k(t-s)} \;\text{ for } t \ge s, \qquad G(t,s) = 0 \;\text{ for } t < s, \tag{0.8.8}$$

is a **Green's function** — the response at time t to a unit impulse at time s, with the second clause enforcing that causes precede effects. Chapter 0.9 will give the impulse a name (δ), prove that (0.8.7) is a *convolution*, and show that convolutions become ordinary products under Fourier transform — at which point solving linear differential equations becomes division. Everything in this book that computes a response to a disturbance, up to and including the propagators of Chapter 5.6, is (0.8.7) with more indices.

▼ Grind box — checking the solution, and one nonlinear equation solved exactly

Verification. Never trust a derived formula you have not differentiated back. Differentiate (0.8.7), using the fundamental theorem of calculus on the upper limit *and* the chain rule on the t inside the integrand:

$$\begin{aligned}\dot y &= -k\,y(0)\mathrm{e}^{-kt} + \underbrace{\mathrm{e}^{-k(t-t)} f(t)}_{\text{upper limit}} \;-\; k\int_0^t \mathrm{e}^{-k(t-s)} f(s)\,\mathrm{d}s \\ &= f(t) - k\left[y(0)\mathrm{e}^{-kt} + \int_0^t \mathrm{e}^{-k(t-s)} f(s)\mathrm{d}s\right] \;=\; f(t) - k\,y.\end{aligned}$$

So $\dot y + ky = f$, and $y(0) = y(0)$ because the integral is empty at $t = 0$. ✓

A constant infusion. Put $f(t) = R$, a steady rate. Then $\int_0^t \mathrm{e}^{-k(t-s)} R\,\mathrm{d}s = \frac{R}{k}\left(1 - \mathrm{e}^{-kt}\right)$, so

$$y(t) = y(0)\mathrm{e}^{-kt} + \frac{R}{k}\left(1 - \mathrm{e}^{-kt}\right) \;\longrightarrow\; \frac{R}{k}.$$

The steady state is rate in over rate out, approached with time constant $1/k$ — so the time to reach steady state is set by the *elimination* constant and not by the infusion rate, which sets only the level. This is why a loading dose exists as a concept.

A nonlinear equation that still yields. Separation does not care about linearity. The logistic equation $\dot{y} = ry(1 - y/K)$ separates:

$$\int \frac{\mathrm{d}y}{y(1-y/K)} = \int r\,\mathrm{d}t \quad \Longrightarrow \quad \ln\frac{y}{K-y} = rt + C,$$

using the partial fractions $\frac{1}{y(1-y/K)} = \frac{1}{y} + \frac{1/K}{1-y/K}$. Solving for y gives $y(t) = K/\big(1 + A\mathrm{e}^{-rt}\big)$ with $A = (K - y_0)/y_0$. Note what we did *not* get: any right to add two solutions. Superposition is a linear privilege, and §8 is about how much that costs.

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3 · A linear ODE is an eigenvalue problem

This section is the one to read twice. It contains no computation you could not do without it, and it changes what all the computations mean.

3.1 · The solution set is a vector space, and its dimension is the order

Take the homogeneous equation $L[y] = 0$ with L as in (0.8.1). Because L is linear, if $L[y_1] = 0$ and $L[y_2] = 0$ then

$$L[\alpha y_1 + \beta y_2] \;=\; \alpha L[y_1] + \beta L[y_2] \;=\; 0. \tag{0.8.9}$$

Any combination of solutions is a solution, and $y \equiv 0$ is a solution. Chapter 0.4 already identified this: **the solution set of a homogeneous linear ODE is a vector space** — the kernel of the linear map L. That is the structural reason superposition works, and it is worth noticing that it has nothing to do with the physics of waves and everything to do with the algebra of L.

Now the dimension, which is where the quoted theorem of §1 gets spent. Consider the map that takes a solution to its initial data:

$$\Phi:\; y \;\longmapsto\; \big(y(t_0),\, \dot{y}(t_0),\, \ldots,\, y^{(n-1)}(t_0)\big) \;\in\; \mathbb{R}^n. \tag{0.8.10}$$

Φ is linear, because evaluating a derivative at a point is a linear operation. It is *injective*: if two solutions share all n initial values, then by the uniqueness half of Picard–Lindelöf they are the same solution, so $\Phi(y) = 0$ forces $y = 0$. It is *surjective*: given any list of n numbers, the existence half produces a solution with those initial values. So Φ is an isomorphism of vector spaces, and by Chapter 0.4 isomorphic spaces have equal dimension:

$$\boxed{\dim\big(\text{solutions of } L[y] = 0\big) \;=\; n \;=\; \text{the order.}} \tag{0.8.11}$$

Read what has happened to a phrase you have used since school. "The general solution contains two arbitrary constants" is a vague and slightly apologetic statement. **"The solution space is two-dimensional"** is a precise one, and it tells you exactly what to do: *find a basis*. Two independent solutions is not a lucky haul, it is a complete answer, and the "arbitrary constants" are *coordinates* in that basis. Chapter 0.4 proved this for $\ddot{y} + \omega^2 y = 0$ by hand; we now have it for every linear ODE at once.

3.2 · The exponential is the eigenfunction of the derivative

Here is the punchline. Let $D = \mathrm{d}/\mathrm{d}t$, regarded as a linear map on the space of smooth functions — an entirely legitimate vector space, as Chapter 0.4 insisted. Ask Chapter 0.5's question of it: *which vectors does this map merely rescale?* That is

$$Dy \;=\; \lambda y, \qquad \text{i.e.} \qquad \dot{y} = \lambda y, \tag{0.8.12}$$

which we solved in §2: $y = C\mathrm{e}^{\lambda t}$, and only that. So

> **THE SENTENCE THIS CHAPTER IS BUILT ON**
>
> $\mathrm{e}^{\lambda t}$ **is the eigenfunction of** $\mathrm{d}/\mathrm{d}t$**, with eigenvalue** λ**.** Every eigenvalue $\lambda \in \mathbb{C}$ occurs, each with a one-dimensional eigenspace.
>
> Therefore: *solving a constant-coefficient linear ODE is diagonalising the derivative operator.* The reason you were told to "try $\mathrm{e}^{\lambda t}$" is not that somebody once noticed it works. It is that you are looking for the basis in which D is a diagonal matrix, and (0.8.12) says what that basis is.
>
> This recurs, undisguised, three more times. In Chapter 0.9 the eigenfunctions of $\mathrm{d}/\mathrm{d}x$ with purely imaginary eigenvalue are e^{ikx}, and expanding a function in them is the **Fourier transform** — Fourier analysis is diagonalisation of the derivative. In Chapter 4.5 the eigenvectors of $\hat{H}$ are the stationary states, and the reason they are the useful basis is (0.8.12) applied to $\mathrm{i}\hbar\,\partial_t\,|\psi\rangle = \hat{H}\,|\psi\rangle$. In Chapter 5.3 the eigenfunctions of the field's wave operator are the modes, and each one turns out to be an independent oscillator, which is what makes quantising a field possible at all.

3.3 · The characteristic equation, from the eigenvalue view

Now specialise to **constant coefficients**, which is the case where diagonalising D actually solves the problem. Write the equation using D:

$$L[y] \;=\; \big(D^n + a_{n-1}D^{n-1} + \cdots + a_1 D + a_0\big)y \;\equiv\; p(D)\,y, \tag{0.8.13}$$

where p is an ordinary polynomial with the constants as coefficients. Feed an eigenfunction in. Since $D\mathrm{e}^{\lambda t} = \lambda\mathrm{e}^{\lambda t}$, applying D again gives $D^2\mathrm{e}^{\lambda t} = \lambda^2\mathrm{e}^{\lambda t}$, and inductively $D^m\mathrm{e}^{\lambda t} = \lambda^m\mathrm{e}^{\lambda t}$. So *every* term of $p(D)$ acts on $\mathrm{e}^{\lambda t}$ by multiplication, and

$$p(D)\,\mathrm{e}^{\lambda t} \;=\; p(\lambda)\,\mathrm{e}^{\lambda t}. \tag{0.8.14}$$

An exponential is a solution of $p(D)y = 0$ exactly when $p(\lambda) = 0$. That is the **characteristic equation**, and it did not have to be introduced as a recipe: it is (0.8.14), which is Chapter 0.5's statement that a polynomial in an operator acts on an eigenvector by the same polynomial in the eigenvalue. Compare (0.8.15) below with the matrix characteristic polynomial of Chapter 0.5 and you are looking at the same object.

$$\lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1\lambda + a_0 \;=\; 0. \tag{0.8.15}$$

If this degree-n polynomial has n *distinct* roots $\lambda_1, \ldots, \lambda_n$ — over $\mathbb{C}$ it has n roots counted with multiplicity, by the fact recorded in Chapter 0.4 — then the n functions $\mathrm{e}^{\lambda_i t}$ are independent, and by (0.8.11) they are a basis. Done: the general solution is $y = \sum_i c_i \mathrm{e}^{\lambda_i t}$.

Real equations, complex roots. Physics equations have real coefficients, so complex roots come in conjugate pairs $\lambda = a \pm \mathrm{i}b$. The corresponding solutions $\mathrm{e}^{at}\mathrm{e}^{\pm \mathrm{i}bt}$ are complex, but their combinations

$$\frac{\mathrm{e}^{\lambda t} + \mathrm{e}^{\bar\lambda t}}{2} = \mathrm{e}^{at}\cos bt, \qquad \frac{\mathrm{e}^{\lambda t} - \mathrm{e}^{\bar\lambda t}}{2\mathrm{i}} = \mathrm{e}^{at}\sin bt, \tag{0.8.16}$$

are real, by Euler's formula from Chapter 0.3. Since they span the same two-dimensional space, you may use either basis. **A complex eigenvalue with non-zero imaginary part means oscillation; its real part is growth or decay.** That single sentence classifies the behaviour of every linear system in this book, and §5 is nothing but it applied once.

3.4 · Repeated roots: the defective case, and where $t\mathrm{e}^{\lambda t}$ comes from

Suppose p has a double root: $p(\lambda) = (\lambda - \lambda_0)^2 q(\lambda)$ with $q(\lambda_0) \neq 0$. Then $\mathrm{e}^{\lambda_0 t}$ is a solution, but it is only *one* solution, and (0.8.11) insists there must be two associated with this root. The eigenvector supply has run short.

You have seen this before, and it was flagged as a warning. Chapter 0.5's warn box exhibited

$$J = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \tag{0.8.17}$$

a 2×2 matrix with a repeated eigenvalue and only a one-dimensional eigenspace — a **defective** matrix, with no basis of eigenvectors and no diagonal form. The correspondence about to appear is exact, not an analogy.

Write $N = D - \lambda_0$, and restrict attention to the two-dimensional space $V = \ker\big((D - \lambda_0)^2\big)$ of solutions belonging to this root. On V we have $N^2 = 0$ by construction, and $N \neq 0$ (or the eigenspace would be two-dimensional and $\mathrm{e}^{\lambda_0 t}$ would not be alone). An operator with $N^2 = 0 \neq N$ is **nilpotent**, and it cannot be diagonalised: a diagonal N with $N^2 = 0$ would have all diagonal entries zero, hence be 0.

What does V contain? Take any $w \in V$ with $Nw \neq 0$. Then $N(Nw) = N^2 w = 0$, so Nw *is* an eigenvector — it is a multiple of $\mathrm{e}^{\lambda_0 t}$. So we need a function that N maps to $\mathrm{e}^{\lambda_0 t}$ rather than to zero. Try $w = t\mathrm{e}^{\lambda_0 t}$ and compute, with the product rule:

$$N\big(t\mathrm{e}^{\lambda_0 t}\big) = \frac{\mathrm{d}}{\mathrm{d}t}\big(t\mathrm{e}^{\lambda_0 t}\big) - \lambda_0 t\mathrm{e}^{\lambda_0 t} = \mathrm{e}^{\lambda_0 t} + \lambda_0 t\mathrm{e}^{\lambda_0 t} - \lambda_0 t\mathrm{e}^{\lambda_0 t} = \mathrm{e}^{\lambda_0 t}. \tag{0.8.18}$$

Exactly what was needed. Applying N once more kills it, so $(D-\lambda_0)^2\left(te^{\lambda_0 t}\right)=0$ and $te^{\lambda_0 t}$ is the second solution — *derived*, not guessed. And in the basis $\{u,w\}=\{\mathrm{e}^{\lambda_0 t},\ te^{\lambda_0 t}\}$ the operator D acts by $Du=\lambda_0 u$ and $Dw=\lambda_0 w+u$, whose matrix is

$$\begin{pmatrix}\lambda_0 & 1\\ 0 & \lambda_0\end{pmatrix}, \qquad \text{which is exactly } J \text{ when } \lambda_0=1. \tag{0.8.19}$$

So (0.8.17) is not a curiosity a linear-algebra text invented to spoil the spectral theorem. It is what the derivative operator looks like at a repeated root, and the extra factor of t in $te^{\lambda_0 t}$ is the off-diagonal 1. The vector w has a name in linear algebra — a **generalised eigenvector** — and now you know what one is: a function that the operator almost, but not quite, merely rescales.

▼ **Grind box — higher multiplicity, reduction of order, and the confluent limit**

Multiplicity m. The same argument iterates. Since $N\left(t^j\mathrm{e}^{\lambda_0 t}\right)=j\,t^{j-1}\mathrm{e}^{\lambda_0 t}$ by the identical product-rule computation, applying N repeatedly walks down the ladder $t^{m-1}\to t^{m-2}\to\cdots\to 1\to 0$. Hence N^m annihilates $t^j\mathrm{e}^{\lambda_0 t}$ for every $j<m$, and

$$\mathrm{e}^{\lambda_0 t},\ te^{\lambda_0 t},\ t^2\mathrm{e}^{\lambda_0 t},\ \ldots,\ t^{m-1}\mathrm{e}^{\lambda_0 t}$$

are m independent solutions — independent because a vanishing combination would make a polynomial of degree $<m$ identically zero. That is exactly the m the dimension count demands, and the matrix of D on this space is an $m\times m$ Jordan block.

Reduction of order, if you prefer no operators. Knowing one solution $y_1=\mathrm{e}^{\lambda_0 t}$ of $\ddot y-2\lambda_0\dot y+\lambda_0^2 y=0$, substitute $y=v(t)\,\mathrm{e}^{\lambda_0 t}$. Then $\dot y=(\dot v+\lambda_0 v)\mathrm{e}^{\lambda_0 t}$ and $\ddot y=(\ddot v+2\lambda_0\dot v+\lambda_0^2 v)\mathrm{e}^{\lambda_0 t}$, and substituting, every term without a derivative of v cancels:

$$\left(\ddot v+2\lambda_0\dot v+\lambda_0^2 v\right)-2\lambda_0\left(\dot v+\lambda_0 v\right)+\lambda_0^2 v=\ddot v=0.$$

So $v=A+Bt$, recovering $y=(A+Bt)\mathrm{e}^{\lambda_0 t}$. The cancellation is not luck; it is (0.8.14) saying that the coefficient of v is $p(\lambda_0)$ and the coefficient of $\dot v$ is $p'(\lambda_0)$ — both zero at a double root.

The confluent limit, which is the most physical derivation. Take two distinct roots $\lambda_1\neq\lambda_2$. The pair $\{\mathrm{e}^{\lambda_1 t},\mathrm{e}^{\lambda_2 t}\}$ is a basis, but so is the pair $\{\mathrm{e}^{\lambda_1 t},\ (\mathrm{e}^{\lambda_2 t}-\mathrm{e}^{\lambda_1 t})/(\lambda_2-\lambda_1)\}$, since the second is a combination of the first two with non-zero coefficient. Now let $\lambda_2\to\lambda_1$. The second basis vector is a difference quotient in λ, and by the definition of the derivative (Chapter 0.1),

$$\frac{\mathrm{e}^{\lambda_2 t}-\mathrm{e}^{\lambda_1 t}}{\lambda_2-\lambda_1}\ \longrightarrow\ \frac{\partial}{\partial\lambda}\mathrm{e}^{\lambda t}\bigg|_{\lambda_1}=t\,\mathrm{e}^{\lambda_1 t}.$$

The second solution does not appear from nowhere when the roots collide — it is the *derivative with respect to the eigenvalue*, and it was there all along as a finite-difference of the two exponentials. This is the same manoeuvre that will produce the secular term $t\sin\omega_0 t$ at exact resonance in §6, and for the same reason: two things became equal and the difference between them had to be rescaled to survive.

4 · The harmonic oscillator

A mass m on a spring of stiffness k, with no friction and no driving. Hooke's law says the restoring force is $-kx$, so Newton gives $m\ddot{x} = -kx$, i.e.

$$\ddot{x} + \omega^2 x = 0, \qquad \omega \equiv \sqrt{k/m}. \tag{0.8.20}$$

By Chapter 0.3 this is not a statement about springs. Expand any potential V about a stable minimum at x_0: the constant is irrelevant, the linear term vanishes because $V'(x_0) = 0$, and what survives is $V \approx V(x_0) + \frac{1}{2}V''(x_0)(x - x_0)^2$ — a spring with $k = V''(x_0)$. So (0.8.20) governs a pendulum, a diatomic molecule, an atom in a crystal, a mode of the electromagnetic field, and a small perturbation of a black hole. Solve it once.

4.1 · First way: real exponentials that are not real

Apply §3. The characteristic equation is $\lambda^2 + \omega^2 = 0$, so $\lambda = \pm\mathrm{i}\omega$ — a conjugate pair with zero real part. By (0.8.16) a real basis is $\{\cos\omega t,\ \sin\omega t\}$, and

$$x(t) = C_1\cos\omega t + C_2\sin\omega t \;=\; A\cos(\omega t + \varphi). \tag{0.8.21}$$

The second form is the first, rewritten: expanding $A\cos(\omega t + \varphi) = A\cos\varphi\cos\omega t - A\sin\varphi\sin\omega t$ identifies $C_1 = A\cos\varphi$ and $C_2 = -A\sin\varphi$, hence

$$A = \sqrt{C_1^2 + C_2^2}, \qquad \tan\varphi = -C_2/C_1. \tag{0.8.22}$$

Two constants either way, as (0.8.11) demands — the same two-dimensional space in two coordinate systems. Fitting initial conditions $x(0) = x_0$, $\dot{x}(0) = v_0$ gives $C_1 = x_0$ and $C_2 = v_0/\omega$. Note what is *not* in (0.8.21): the frequency does not depend on the amplitude. A pendulum clock keeps time whether it swings wide or narrow, and §8 is about the fact that this is only approximately true.

4.2 · Second way: the complex amplitude

Physicists rarely write (0.8.21). They write

$$x(t) \;=\; \mathrm{Re}\big[Z\,\mathrm{e}^{\mathrm{i}\omega t}\big], \qquad Z = A\,\mathrm{e}^{\mathrm{i}\varphi} \in \mathbb{C}, \tag{0.8.23}$$

which is the same thing: $\mathrm{Re}[A\mathrm{e}^{\mathrm{i}\varphi}\mathrm{e}^{\mathrm{i}\omega t}] = A\cos(\omega t+\varphi)$ by Euler. The gain is bookkeeping, and it is large. **The single complex number Z carries both the amplitude (its modulus) and the phase (its argument).** Adding two oscillations of the same frequency becomes adding two complex numbers; shifting a phase becomes multiplying by $\mathrm{e}^{\mathrm{i}\theta}$, which Chapter 0.3 showed is a rotation. Differentiating becomes multiplying by $\mathrm{i}\omega$ — the derivative has been turned into an algebraic operation, which is exactly (0.8.12) being used as a labour-saving device.

The move is legal for one reason worth stating: the equation is linear with *real* coefficients, so if $z(t)$ solves it then so do $\bar{z}$ and hence $\mathrm{Re}\, z = (z+\bar{z})/2$. You may solve the complex problem and take the real part at the end. The moment the equation is nonlinear this fails — $\mathrm{Re}(z^2) \neq (\mathrm{Re}\, z)^2$ — which is one more entry on §8's list.

4.3 · Third way: energy, and the first integral

The third route does not solve the equation at all; it reduces its order, and it generalises further than either of the others. Multiply (0.8.20) by $m\dot{x}$:

$$m\dot{x}\ddot{x} + m\omega^2 x\dot{x} = 0 \quad\Longrightarrow\quad \frac{\mathrm{d}}{\mathrm{d}t}\left[\tfrac{1}{2}m\dot{x}^2 + \tfrac{1}{2}m\omega^2x^2\right] = 0, \tag{0.8.24}$$

because each term is the derivative of a square by the chain rule. So the bracket is constant:

$$E \;=\; \tfrac{1}{2}m\dot{x}^2 + \tfrac{1}{2}kx^2 \;=\; \text{constant}, \qquad k = m\omega^2. \tag{0.8.25}$$

A quantity that is constant along solutions is a **first integral**, and finding one converts a second-order equation into a first-order one. Here it does more: solve (0.8.25) for $\dot{x}$ and you get a separable equation whose integration reproduces (0.8.21). Chapter 1.4 will explain where first integrals come from in general — every one of them is a symmetry, and this one is invariance under shifting the clock.

4.4 · Phase space: the trajectory is a closed ellipse

Plot the state (x,p) with $p = m\dot{x}$ rather than plotting x against t. Rewriting (0.8.25) in these variables,

$$\frac{p^2}{2mE} + \frac{x^2}{2E/(m\omega^2)} \;=\; 1, \tag{0.8.26}$$

an **ellipse** with semi-axes $x_{\max} = A = \sqrt{2E/(m\omega^2)}$ and $p_{\max} = m\omega A$. Every solution runs around it, clockwise, at uniform angular rate in the rescaled coordinates $(\omega x,\ \dot{x})$ — where the ellipse becomes a circle and the motion becomes literally a rotation, which is (0.8.23) drawn as a picture.

That the curve *closes* is the whole content of energy conservation. An orbit that failed to close would return to the same position with a different speed, and E would have changed. So "the trajectory is a closed loop in phase space" and "energy is conserved" are the same statement, one drawn and one written.

Compute the enclosed area, since the answer will matter later. The area of an ellipse is π times the product of the semi-axes:

$$\mathcal{A} = \pi\, x_{\max}\, p_{\max} = \pi\, A \cdot m\omega A = \pi m\omega A^2 = \frac{2\pi E}{\omega} = E\,T, \qquad (0.8.27)$$

using $E = \frac{1}{2}m\omega^2 A^2$ and $T = 2\pi/\omega$. **Phase-space area is energy times period.** Hold onto that, because Chapter 4.5 will find that a quantum oscillator has $E_n = (n + \frac{1}{2})\hbar\omega$, whence

$$\mathcal{A}_n = \frac{2\pi E_n}{\omega} = 2\pi\, \left(n + \tfrac{1}{2}\right) \hbar = \left(n + \tfrac{1}{2}\right) h. \qquad (0.8.28)$$

The allowed orbits are those enclosing an integer-and-a-half multiple of Planck's constant. The classical ellipse is not replaced in quantum mechanics; it is *rationed*. Chapter 4.5 derives this properly with ladder operators, and Chapter 1.3 will show that phase-space area is preserved by all of classical mechanics, not just this example — which is why it was the right thing to quantise.

5 · Damping

Nothing oscillates forever. Add a resistive force proportional to velocity — viscous drag, an electrical resistance, the radiation of energy away to infinity — and Newton's law reads $m\ddot{x} = -kx - b\dot{x}$. Divide by m and name the two constants:

$$\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = 0, \qquad \gamma \equiv \frac{b}{2m}, \quad \omega_0 \equiv \sqrt{k/m}. \qquad (0.8.29)$$

The factor of 2 in the definition of γ is chosen purely to make what follows tidy, and you should expect it to be worth its keep. Section 3 does all the work: the characteristic equation is $\lambda^2 + 2\gamma\lambda + \omega_0^2 = 0$, so

$$\lambda_\pm = -\gamma \pm \sqrt{\gamma^2 - \omega_0^2}. \qquad (0.8.30)$$

Everything about damped motion is in the discriminant $\gamma^2 - \omega_0^2$, and there are exactly three cases because a real number is negative, zero, or positive.

5.1 · Underdamped: $\gamma < \omega_0$

The discriminant is negative, so the roots are a complex conjugate pair $\lambda_\pm = -\gamma \pm \mathrm{i}\omega_\mathrm{d}$ with

$$\omega_\mathrm{d} = \sqrt{\omega_0^2 - \gamma^2}, \qquad x(t) = A\,\mathrm{e}^{-\gamma t}\cos(\omega_\mathrm{d} t + \varphi)\,. \qquad (0.8.31)$$

Read it off (0.8.16): the real part $-\gamma$ is the decay, the imaginary part ω_d is the oscillation. The system rings and the ringing dies away with time constant $1/\gamma$. Two details are worth noticing. First, damping *lowers* the frequency, $\omega_\mathrm{d} < \omega_0$ — the system is dragged, so it takes longer to come back. Second, the effect is second-order small: $\omega_\mathrm{d} \approx \omega_0\big(1 -$

$\frac{1}{2}\gamma^2/\omega_0^2)$, so a system that loses a per cent of its amplitude per radian shifts its frequency by only a part in 2×10^4. Weak damping kills the amplitude long before it moves the pitch.

5.2 · Overdamped: $\gamma > \omega_0$

The discriminant is positive, both roots are real and negative, and the motion is a sum of two decaying exponentials with no oscillation at all:

$$x(t) = C_+ \mathrm{e}^{\lambda_+ t} + C_- \mathrm{e}^{\lambda_- t}, \qquad \lambda_\pm = -\gamma \pm \sqrt{\gamma^2 - \omega_0^2} < 0. \tag{0.8.32}$$

Both are negative because $\sqrt{\gamma^2 - \omega_0^2} < \gamma$. At long times the slower one — λ_+, the one closer to zero — dominates, and here is the counter-intuitive part. For heavy damping, $\gamma \gg \omega_0$, expand the square root by Chapter 0.3:

$$\lambda_+ = -\gamma \left(1 - \sqrt{1 - \omega_0^2/\gamma^2}\right) \approx -\frac{\omega_0^2}{2\gamma}. \tag{0.8.33}$$

More damping makes the return to equilibrium slower, not faster. A door closer filled with treacle takes forever. The rate is inversely proportional to the drag, exactly as in the overdamped limit of any relaxation problem.

5.3 · Critically damped: $\gamma = \omega_0$

The discriminant vanishes and $\lambda = -\gamma$ is a *repeated* root. This is precisely the defective case of §3.4, and the second solution is forced to be (0.8.18):

$$x(t) = (A + Bt)\,\mathrm{e}^{-\gamma t}. \tag{0.8.34}$$

So the extra factor of t in a critically damped shock absorber and the off-diagonal 1 in the Jordan block (0.8.17) are the same fact. Critical damping is the boundary between ringing and crawling, and it is the fastest possible approach to equilibrium without overshoot — which is why car suspensions, galvanometers and control loops are tuned to it. That it is fastest can be read off (0.8.33): the decay exponent is $\gamma = \omega_0$ here, larger in magnitude than the $\omega_0^2/2\gamma$ available on the overdamped side and larger than the envelope $\mathrm{e}^{-\gamma t}$ available on the underdamped side, since $\gamma < \omega_0$ there.

5.4 · The quality factor

The dimensionless combination that controls everything is

$$\boxed{Q \;\equiv\; \frac{\omega_0}{2\gamma}} \tag{0.8.35}$$

and its meaning is energetic. Energy in an oscillator is quadratic in amplitude, and the amplitude envelope decays as $\mathrm{e}^{-\gamma t}$, so

$$E(t) = E_0\,\mathrm{e}^{-2\gamma t} = E_0\,\mathrm{e}^{-\omega_0 t/Q}. \tag{0.8.36}$$

In time t the oscillator has advanced $\omega_0 t$ radians of phase, and the exponent is that number divided by Q. Hence, exactly:

Q is the number of radians of oscillation in which the energy falls by a factor of e.

Divide by 2π and it is the number of *cycles*, times 2π; a system with $Q = 100$ rings for about 16 cycles before losing 63% of its energy. The often-quoted statement "the fractional energy lost per cycle is $2\pi/Q$" is the small-loss expansion of (0.8.36): over one period $T = 2\pi/\omega_0$ the energy ratio is $\mathrm{e}^{-2\pi/Q} \approx 1 - 2\pi/Q$, which is accurate to a per cent only for $Q \gtrsim 300$. Use the exponential; it is no harder.

Section 6 will produce a second, entirely independent meaning for the same number — Q is the sharpness of the resonance peak — and the fact that one number does both jobs is the width–lifetime relation that runs through the rest of this book.

▼ **Grind box — all three cases with the same initial conditions, and the seam between them**

Release from rest: $x(0) = x_0$, $\dot{x}(0) = 0$. Impose these on each case.

Underdamped. From (0.8.31) written as $x = \mathrm{e}^{-\gamma t}(A\cos\omega_\mathrm{d} t + B\sin\omega_\mathrm{d} t)$: at $t = 0$, $A = x_0$; differentiating, $\dot{x}(0) = -\gamma A + \omega_\mathrm{d} B = 0$ gives $B = \gamma x_0/\omega_\mathrm{d}$, so

$$x(t) = x_0\,\mathrm{e}^{-\gamma t}\left[\cos\omega_\mathrm{d} t + \frac{\gamma}{\omega_\mathrm{d}}\sin\omega_\mathrm{d} t\right].$$

Critically damped. From (0.8.34): $A = x_0$ and $\dot{x}(0) = B - \gamma A = 0$ gives $B = \gamma x_0$, so $x(t) = x_0\mathrm{e}^{-\gamma t}(1 + \gamma t)$.

The seam is continuous. Let $\omega_\mathrm{d} \to 0$ in the underdamped answer. Then $\cos\omega_\mathrm{d} t \to 1$ and $\sin(\omega_\mathrm{d} t)/\omega_\mathrm{d} \to t$, so the bracket tends to $1 + \gamma t$ — exactly the critical solution. Nothing discontinuous happens at $\gamma = \omega_0$; the oscillation simply has its first zero crossing pushed out to $t = \pi/\omega_\mathrm{d} \to \infty$. This is the confluent limit of the grind box in §3 seen in physical clothing, and it is why the repeated-root case is not a separate phenomenon.

Overdamped. Same algebra with $\omega_\mathrm{d} \to \mathrm{i}\kappa$, $\kappa = \sqrt{\gamma^2 - \omega_0^2}$, turns the trigonometric functions into hyperbolic ones:

$$x(t) = x_0\,\mathrm{e}^{-\gamma t}\left[\cosh\kappa t + \frac{\gamma}{\kappa}\sinh\kappa t\right],$$

which is positive for all $t > 0$: an overdamped system released from rest never crosses zero. The critically damped case is the last one for which that is true, and the underdamped case crosses first at $\omega_\mathrm{d} t = \pi - \arctan(\gamma/\omega_\mathrm{d})$.

6 · Driving, resonance, and the shape of every spectral line

Now push on it. Take a sinusoidal drive of angular frequency ω, which need not be ω_0, and write the force per unit mass as $F_0 \cos\omega t$:

$$\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = F_0 \cos\omega t. \tag{0.8.37}$$

This is inhomogeneous, and Chapter 0.4 already told us the structure of the answer: *general solution = one particular solution + the whole vector space of homogeneous solutions*. The homogeneous part is §5, and it decays as $\mathrm{e}^{-\gamma t}$. So after a few times $1/\gamma$ the system forgets how it was started and settles onto the particular solution. That survivor is the **steady state**, and — this will matter in a moment — *it exists as a well-defined attractor only because* $\gamma > 0$.

6.1 · Solving it with a complex amplitude

Solve the complex equation $\ddot{z} + 2\gamma\dot{z} + \omega_0^2 z = F_0\mathrm{e}^{\mathrm{i}\omega t}$ and take the real part at the end, which is legal because (0.8.37) is linear with real coefficients. Try $z = Z\mathrm{e}^{\mathrm{i}\omega t}$ with $Z \in \mathbb{C}$ constant. Each derivative brings down a factor $\mathrm{i}\omega$ — this is (0.8.12) earning its keep, since $\mathrm{e}^{\mathrm{i}\omega t}$ is an eigenfunction of D — so the differential equation collapses to an algebraic one:

$$\left(-\omega^2 + 2\mathrm{i}\gamma\omega + \omega_0^2\right) Z = F_0 \qquad \Longrightarrow \qquad Z = \frac{F_0}{\omega_0^2 - \omega^2 + 2\mathrm{i}\gamma\omega}. \tag{0.8.38}$$

That is the entire solution. Writing $Z = A\mathrm{e}^{-\mathrm{i}\delta}$ so that $x(t) = A\cos(\omega t - \delta)$ — with δ the amount by which the response *lags* the drive — and taking modulus and argument of (0.8.38):

$$\begin{aligned} A(\omega) &= \frac{F_0}{\sqrt{\left(\omega_0^2 - \omega^2\right)^2 + 4\gamma^2\omega^2}}, \\ \tan\delta(\omega) &= \frac{2\gamma\omega}{\omega_0^2 - \omega^2}, \qquad \delta \in (0, \pi). \end{aligned} \tag{0.8.39}$$

The branch is fixed by requiring δ to be continuous and to start at 0: the imaginary part of the denominator of Z is $2\gamma\omega > 0$, so δ is always in the upper half range.

6.2 · Where the amplitude peaks — and it is not at ω_0

Maximise A by minimising the quantity under the root. Write $s = \omega^2$, so we are minimising $g(s) = (\omega_0^2 - s)^2 + 4\gamma^2 s$; then $g'(s) = -2(\omega_0^2 - s) + 4\gamma^2 = 0$ gives $s = \omega_0^2 - 2\gamma^2$, i.e.

$$\boxed{\omega_{\text{peak}} = \sqrt{\omega_0^2 - 2\gamma^2}} \qquad A_{\max} = \frac{F_0}{2\gamma\sqrt{\omega_0^2 - \gamma^2}}, \tag{0.8.40}$$

the amplitude following by substituting $\omega_0^2 - \omega_{\text{peak}}^2 = 2\gamma^2$ back into (0.8.39). Three things fall out. The peak is at neither the natural frequency ω_0 nor the damped frequency $\omega_{\mathrm{d}} = \sqrt{\omega_0^2 - \gamma^2}$ but at a third, still lower value. It exists at all only when $\omega_0^2 > 2\gamma^2$, that is $Q > 1/\sqrt{2}$; below that the response falls monotonically and there is no resonance. And as $\gamma \to 0$ the peak height diverges like $F_0/(2\gamma\omega_0)$ — a fact the warning box below is about.

6.3 · Power absorbed, and the exact width

Amplitude is not the physically cleanest quantity; *power* is, because that is what a spectrometer measures. The instantaneous power delivered by the drive is force times velocity. With $x = A\cos(\omega t - \delta)$, so $\dot{x} = -A\omega\sin(\omega t - \delta)$, average over a cycle using $\langle\cos^2\rangle = \frac{1}{2}$ and $\langle\sin\cos\rangle = 0$:

$$\begin{aligned}\bar{P} &= \Big\langle F_0\cos\omega t\cdot(-A\omega\sin(\omega t-\delta))\Big\rangle \\ &= -F_0 A\omega\Big\langle \cos\omega t\,\sin\omega t\cos\delta - \cos^2\omega t\,\sin\delta\Big\rangle \;=\; \tfrac{1}{2}F_0 A\omega\sin\delta.\end{aligned} \tag{0.8.41}$$

Only the component of the response *in phase with the velocity* absorbs energy — the $\sin\delta$ — which is why phase matters physically and not just bookkeeping-ly. From (0.8.38), $\sin\delta = 2\gamma\omega A/F_0$, so

$$\bar{P}(\omega) \;=\; \gamma\,\omega^2 A^2 \;=\; \frac{\gamma F_0^2\,\omega^2}{\left(\omega_0^2-\omega^2\right)^2 + 4\gamma^2\omega^2}. \tag{0.8.42}$$

Now maximise *this*. Divide numerator and denominator by ω^2:

$$\bar{P} = \frac{\gamma F_0^2}{\left(\dfrac{\omega_0^2-\omega^2}{\omega}\right)^2 + 4\gamma^2}, \tag{0.8.43}$$

whose denominator is smallest exactly when the first bracket vanishes, i.e. at $\omega = \omega_0$ precisely, with $\bar{P}(\omega_0) = F_0^2/4\gamma$. **The power resonance sits exactly at the natural frequency, even though the amplitude resonance does not.** Half-power points are just as clean: setting $\bar{P} = \frac{1}{2}\bar{P}(\omega_0)$ requires the bracket to equal $\pm 2\gamma$,

$$\frac{\omega_0^2-\omega^2}{\omega} = \pm 2\gamma \quad\Longrightarrow\quad \omega^2 \pm 2\gamma\omega - \omega_0^2 = 0 \quad\Longrightarrow\quad \omega_\pm = \sqrt{\omega_0^2+\gamma^2}\;\pm\;\gamma, \tag{0.8.44}$$

taking the positive roots. Subtracting, the square roots cancel and

$$\boxed{\text{FWHM} \;=\; \omega_+ - \omega_- \;=\; 2\gamma} \qquad \text{and} \qquad \omega_+\omega_- = \omega_0^2. \tag{0.8.45}$$

Not approximately 2γ — exactly 2γ, for every γ, with the resonant frequency recoverable as the geometric mean of the half-power points. Combining with (0.8.35),

$$Q = \frac{\omega_0}{2\gamma} = \frac{\omega_0}{\text{FWHM}} = \frac{\text{centre frequency}}{\text{width}}. \tag{0.8.46}$$

The same Q that counted radians of ringdown in (0.8.36) now measures the sharpness of a peak in frequency. One number, two apparently unrelated jobs. That coincidence is not a coincidence, and it is the point of this section.

6.4 · The Lorentzian

Near resonance the exact curve (0.8.42) takes a universal form. Factor $\omega_0^2 - \omega^2 = (\omega_0 - \omega)(\omega_0 + \omega)$ and set $\omega_0 + \omega \approx 2\omega_0, \omega \approx \omega_0$ elsewhere — valid when $|\omega - \omega_0| \ll \omega_0$, i.e. for a sharp resonance. Then

$$\bar{P}(\omega) \;\approx\; \frac{\gamma F_0^2 \omega_0^2}{4\omega_0^2\left[(\omega-\omega_0)^2 + \gamma^2\right]} \;=\; \underbrace{\frac{F_0^2}{4\gamma}}_{\bar{P}(\omega_0)} \cdot \frac{\gamma^2}{(\omega-\omega_0)^2 + \gamma^2}. \tag{0.8.47}$$

That shape — a peak of half-width γ falling off as the inverse square of the detuning — is the **Lorentzian**. It is worth being clear about what has and has not been assumed. Nothing about springs. Only two ingredients entered: *one* mode with a natural frequency ω_0, and a decay rate γ for its amplitude. Any system with those two properties, driven near resonance, has this lineshape.

THIS CURVE IS A PARTICLE

Follow the chain, because it ends somewhere surprising.

Spectral lines. An excited atom is a mode with frequency ω_0 that decays, because it radiates. Its emitted field is $e^{-\gamma t}e^{-i\omega_0 t}$ for $t > 0$. Chapter 0.9 will compute the Fourier transform of exactly this function and get $1/\left[i(\omega_0 - \omega) + \gamma\right]$, whose squared modulus is (0.8.47). So the *natural linewidth* of a spectral line is the Lorentzian, and its width is the decay rate. A line is not infinitely sharp because the state emitting it does not last forever.

Unstable particles. Now the same statement with the same mathematics and different words. In Chapter 5.7 you will scatter two particles and plot the cross-section against centre-of-mass energy, and find a bump. Its shape is the **Breit–Wigner** formula

$$\sigma(E) \;\propto\; \frac{\Gamma^2/4}{(E - E_0)^2 + \Gamma^2/4},$$

which is (0.8.47) with $E = \hbar\omega$ and $\Gamma = 2\hbar\gamma$. The unstable particle is a mode of a quantum field with a natural frequency $E_0/\hbar$ and a decay rate; the beam is the drive; the bump is the resonance curve. **A particle detected as a bump in a cross-section is literally a driven damped oscillator.**

And (0.8.36) becomes a measurement technique. The probability of survival decays as $e^{-2\gamma t} = e^{-t/\tau}$ with $\tau = 1/2\gamma$, while (0.8.45) says the width in angular frequency is 2γ. Multiply:

$$\Gamma\tau = \hbar \qquad \Longleftrightarrow \qquad \text{width} = \frac{\hbar}{\text{lifetime}}.$$

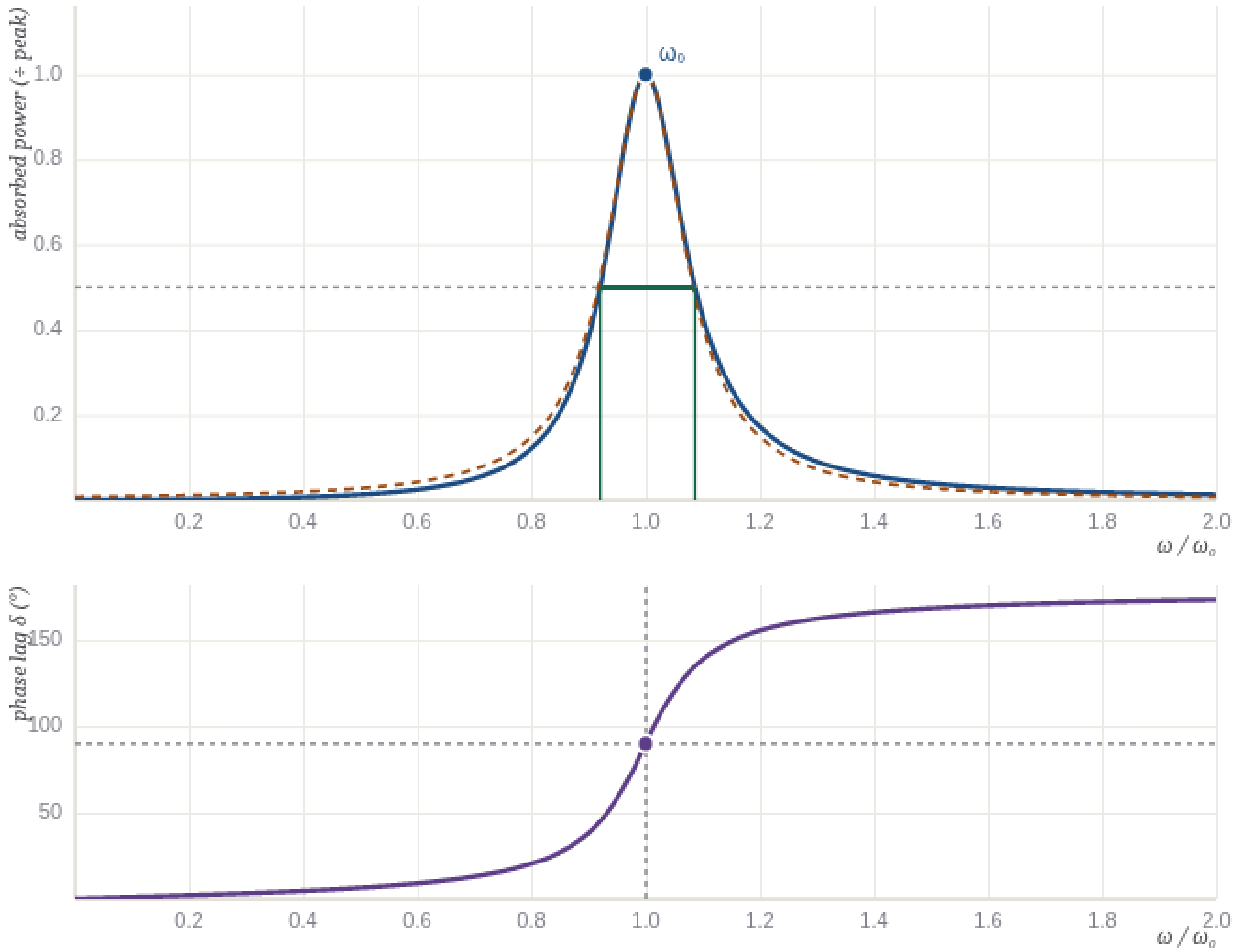
The ρ meson has $\Gamma \approx 150$ MeV; with $\hbar = 6.58 \times 10^{-22}$ MeV s that is a lifetime of 4.4×10^{-24} s. The Z boson has $\Gamma = 2.50$ GeV, hence $\tau = 2.6 \times 10^{-25}$ s. Nobody timed those particles. Nobody could. **They were measured by the width of a peak**, using the identity you just derived from a mass on a spring.

6.5 · The phase, and why its sign structure matters

Look again at $\tan\delta = 2\gamma\omega/(\omega_0^2 - \omega^2)$. Three regimes:

- $\omega \ll \omega_0$: the denominator is positive and large, $\delta \to 0$. The mass follows the force. Push slowly and it goes where you push — the response is *stiffness-controlled*.
- $\omega = \omega_0$: the denominator vanishes and $\delta = \pi/2$ exactly, for any γ whatsoever. The displacement lags the force by a quarter cycle, which puts the *velocity* exactly in phase with the force — which is why (0.8.41) is maximal here, and why the power resonance is exactly at ω_0 while the amplitude resonance is not.
- $\omega \gg \omega_0$: the denominator is negative and large, $\delta \to \pi$. The mass moves *opposite* to the applied force. Shake a heavy object fast and it stays put while your hand goes back and forth around it — the response is *mass-controlled*.

That a resonance flips the sign of the response as you pass through it is one of the most consequential facts in physics. It is why the refractive index of glass falls with wavelength in the visible and rises again beyond an absorption line — *anomalous dispersion* — and the mathematical statement that the real and imaginary parts of a response function (here $\operatorname{Re} Z$ and $\operatorname{Im} Z$) are not independent but determine each other is the Kramers–Kronig relation, which is causality written in the frequency domain. The Green's function of (0.8.8) vanishing for $t < s$ is the same statement in the time domain.



quality Q Q = 6.00 γ = 0.08333

measured FWHM = 0.1666 (2γ = 0.1667)

energy lifetime τ = 1/2γ = 6.00 → FWHM × τ = 1.000

The width is the inverse lifetime. Top: absorbed power (0.8.42), normalised to its peak, in blue; the Lorentzian approximation (0.8.47) dashed in orange. Green marks the measured full width at half maximum, found by scanning the exact curve, not by using the formula — compare it with the printed 2γ and watch them agree to four decimals at every Q. The Lorentzian hugs the peak and departs in the wings, which is exactly what "valid near resonance" means. Bottom: the phase lag, passing through 90° at ω_0 for every Q and tending to 180° above. Slide Q up: the peak narrows and the ringdown lengthens, and the last readout — the product of the width and the energy lifetime — stays pinned at 1. That number is $\Gamma\tau/\hbar$ for the Z boson.

⚠ WHY THIS ISN'T OBVIOUS

Set $\gamma = 0$ and the steady state does not merely become large — it ceases to exist. Put $\gamma = 0$ and $\omega = \omega_0$ in (0.8.39) and you get $A = F_0/0$. The formula has not become large; it has become meaningless, because the assumption behind it — that there is a bounded periodic solution — is false. Here is what actually happens.

For $\omega \neq \omega_0$ the undamped equation $\ddot{x} + \omega_0^2 x = F_0 \cos \omega t$ has the solution starting from rest

$$x(t) = \frac{F_0 (\cos \omega t - \cos \omega_0 t)}{\omega_0^2 - \omega^2},$$

as you can check by differentiating twice. Now let $\omega \to \omega_0$. Numerator and denominator both vanish, so take the limit as a derivative with respect to ω — the confluent move of §3.4 for the third time:

$$x(t) \quad \longrightarrow \quad \frac{-t \sin \omega_0 t}{-2\omega_0} F_0 \quad = \quad \frac{F_0}{2\omega_0} t \sin \omega_0 t.$$

The amplitude grows *linearly and without bound*. There is no steady state to speak of, and the factor of t is the same t as in $te^{\lambda t}$: at exact resonance the driving frequency has collided with a root of the characteristic polynomial, and a repeated root always costs you a factor of t.

This never happens in nature, because $\gamma > 0$ always — every real system leaks energy somewhere. So the honest statement is not "at resonance the amplitude is infinite" but **"the steady state of a driven oscillator exists only because of dissipation, and its peak height $F_0/2\gamma\omega_0$ is set entirely by the loss"**. Resonance is not a property of the oscillator alone. It is a negotiation between the drive and the losses, and the sharper the resonance the more completely the losses are in charge.

7 · Coupled oscillators, normal modes, and the bridge to fields

Everything so far has had one degree of freedom. Almost nothing in nature does. This section adds a second one, discovers that the added complexity is an illusion, and then adds infinitely many.

7.1 · Two masses, three springs

Two equal masses m sit in a line. Each is tied to a wall by a spring of stiffness k, and they are tied to each other by a spring of stiffness κ. Let x_1, x_2 be their displacements from equilibrium. The coupling spring is stretched by $x_1 - x_2$, so it pulls mass 1 back with force $-\kappa(x_1 - x_2)$ and mass 2 with the opposite sign:

$$\begin{aligned} m\ddot{x}_1 &= -kx_1 - \kappa\,(x_1 - x_2)\,, \\ m\ddot{x}_2 &= -kx_2 - \kappa\,(x_2 - x_1)\,. \end{aligned} \qquad (0.8.48)$$

Neither equation can be solved alone: each contains the other's unknown. That is what "coupled" means, and it is why the problem looks harder than §4. Write it as one vector equation with $\mathbf{x} = (x_1, x_2)^\mathsf{T}$:

$$\ddot{\mathbf{x}} = -M\mathbf{x}, \qquad M = \frac{1}{m}\begin{pmatrix} k+\kappa & -\kappa \\ -\kappa & k+\kappa \end{pmatrix}. \tag{0.8.49}$$

M **is symmetric**, and that is not an accident of this example. The forces came from a potential energy

$$V(x_1, x_2) = \tfrac{1}{2}kx_1^2 + \tfrac{1}{2}kx_2^2 + \tfrac{1}{2}\kappa\left(x_1 - x_2\right)^2, \qquad m\ddot{x}_i = -\frac{\partial V}{\partial x_i}, \tag{0.8.50}$$

so $mM_{ij} = \partial_i \partial_j V$ is the **Hessian** of the potential — and Chapter 0.6 proved, via Clairaut, that a Hessian is always symmetric. Any system of masses moving in a potential near a stable equilibrium has this form, because Chapter 0.6's second-order Taylor expansion says $V \approx V_0 + \frac{1}{2}\mathbf{x}^\mathsf{T} H\mathbf{x}$ at a critical point. So the analysis below is not about this apparatus. It is about every stable multi-degree-of-freedom system there is.

7.2 · The spectral theorem does the whole job

By Chapter 0.5, a real symmetric matrix has real eigenvalues and an *orthonormal* basis of eigenvectors. Call them $\mathbf{u}_1, \mathbf{u}_2$ with $M\mathbf{u}_k = \omega_k^2 \mathbf{u}_k$. (Writing the eigenvalue as ω_k^2 presumes it is positive; it is, exactly when the equilibrium is a minimum rather than a saddle — Chapter 0.6's classification, now doing physical work. A negative eigenvalue would give ω imaginary and $\mathrm{e}^{|\omega|t}$ growth: an instability, which is what a saddle *is*.)

Define the **normal coordinates** $q_k = \mathbf{u}_k \cdot \mathbf{x}$, the components of the displacement along the eigenvectors. Differentiate twice and use (0.8.49), remembering that $\mathbf{u}_k$ is a constant vector and that M is symmetric so it can be moved across the dot product:

$$\ddot{q}_k = \mathbf{u}_k \cdot \ddot{\mathbf{x}} = -\mathbf{u}_k \cdot M\mathbf{x} = -\left(M\mathbf{u}_k\right) \cdot \mathbf{x} = -\omega_k^2\, \mathbf{u}_k \cdot \mathbf{x} = -\omega_k^2 q_k. \tag{0.8.51}$$

Look at what that is. Each q_k obeys $\ddot{q}_k = -\omega_k^2 q_k$ — the equation of §4, on its own, with no reference to the other coordinate. **The coupled system has been turned into two independent harmonic oscillators by a rotation of axes.** The step that made it work is the one marked in the middle of (0.8.51), and it needed $M^\mathsf{T} = M$; without symmetry there is no orthonormal eigenbasis and this manoeuvre fails.

7.3 · The two modes, explicitly

Diagonalise M from (0.8.49). It has the form $aI + b\sigma$ with $\sigma = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, whose eigenvectors are $(1,1)$ and $(1,-1)$ by inspection — swap the entries and each is unchanged up to a sign. So

$$\begin{aligned} \mathbf{u}_1 &= \tfrac{1}{\sqrt{2}}(1,1), & \omega_1^2 &= \frac{k}{m}, \\ \mathbf{u}_2 &= \tfrac{1}{\sqrt{2}}(1,-1), & \omega_2^2 &= \frac{k+2\kappa}{m}. \end{aligned} \tag{0.8.52}$$

Verify the eigenvalues directly: $M(1,1)^\mathsf{T} = \frac{1}{m}(k+\kappa-\kappa)(1,1)^\mathsf{T}$ and $M(1,-1)^\mathsf{T} = \frac{1}{m}(k+\kappa+\kappa)(1,-1)^\mathsf{T}$. And note $\mathbf{u}_1 \cdot \mathbf{u}_2 = \frac{1}{2}(1-1) = 0$: perpendicular, as Chapter 0.5 promised, without our having arranged it.

The physical reading is the pleasure of the calculation. In **mode 1** the two masses move together, in phase, with equal amplitude. The coupling spring is then never stretched, so it exerts no force at all, and the frequency is $\sqrt{k/m}$ — *independent of* κ, exactly as if the coupling spring were not there. In **mode 2** they move exactly out of phase. The midpoint stays fixed, the coupling spring is stretched by $2x_1$, and the restoring force on each mass is $-(k+2\kappa)x_1$: stiffer, hence faster. The *only* two ways this system can move sinusoidally are these, and everything else is a superposition of them.

General motion, then, is

$$\mathbf{x}(t) = \sum_{k=1,2} \left[a_k \cos\omega_k t + \frac{b_k}{\omega_k}\sin\omega_k t \right] \mathbf{u}_k, \qquad \begin{aligned} a_k &= \mathbf{u}_k \cdot \mathbf{x}(0), \\ b_k &= \mathbf{u}_k \cdot \dot{\mathbf{x}}(0), \end{aligned} \tag{0.8.53}$$

the coefficients being read off by orthonormality — Chapter 0.5's *coordinates have become inner products*, used here for the first time on a physical problem. Four constants for a four-dimensional solution space (two second-order equations), consistent with (0.8.11).

7.4 · Beats

Now the initial condition that makes the structure visible. Pull mass 1 aside by d, hold mass 2 at rest at its equilibrium, and release both from rest: $\mathbf{x}(0) = (d,0), \dot{\mathbf{x}}(0) = \mathbf{0}$. Then $a_1 = d/\sqrt{2}, a_2 = d/\sqrt{2}, b_k = 0$, and (0.8.53) gives

$$x_1(t) = \tfrac{d}{2}\left(\cos\omega_1 t + \cos\omega_2 t\right), \qquad x_2(t) = \tfrac{d}{2}\left(\cos\omega_1 t - \cos\omega_2 t\right). \tag{0.8.54}$$

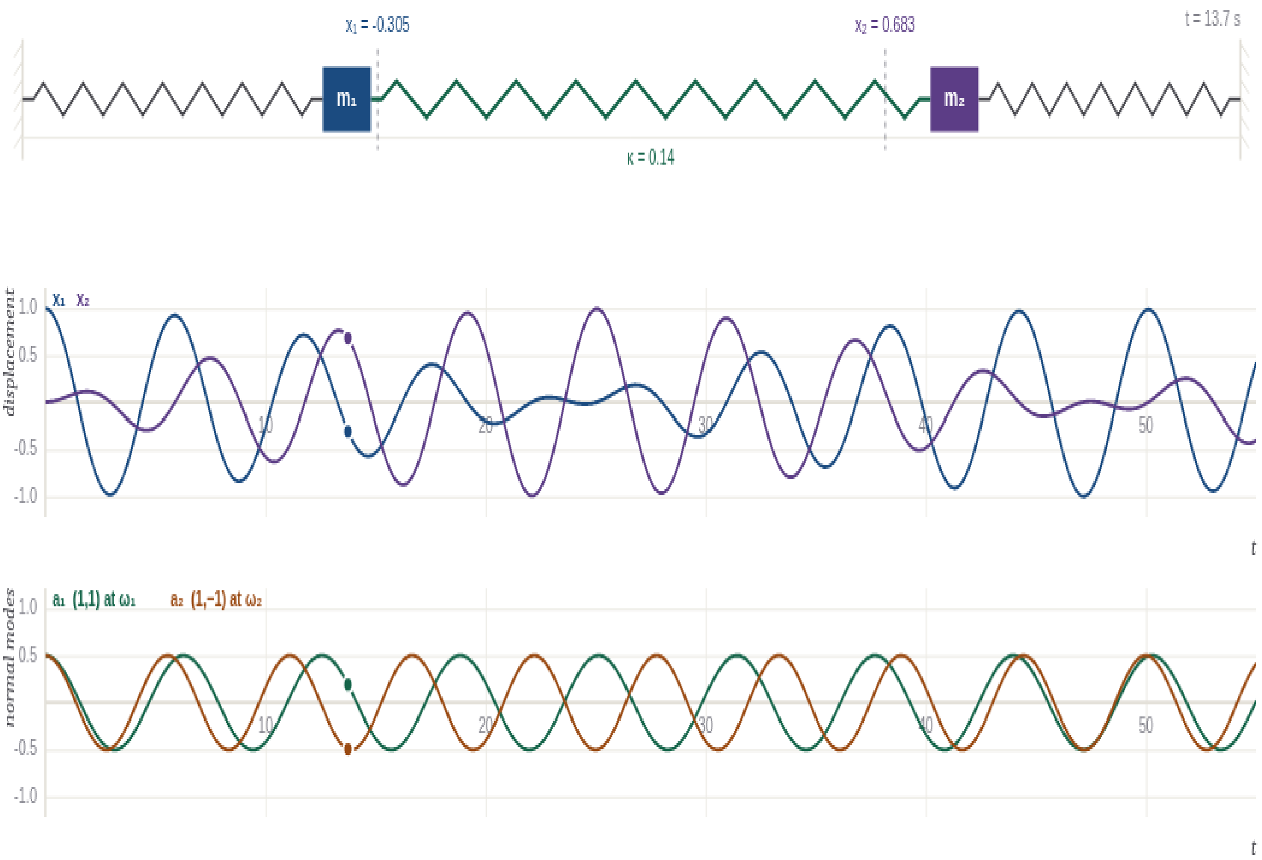
Apply the sum-to-product identities — which are just Euler's formula rearranged — with $\bar{\omega} = \frac{1}{2}(\omega_1+\omega_2)$ and $\Delta = \omega_2 - \omega_1$:

$$x_1(t) = d\,\cos\!\left(\tfrac{\Delta t}{2}\right)\cos\bar{\omega}t, \qquad x_2(t) = d\,\sin\!\left(\tfrac{\Delta t}{2}\right)\sin\bar{\omega}t. \tag{0.8.55}$$

Each mass oscillates fast at the mean frequency $\bar{\omega}$, inside a slowly changing envelope of frequency $\Delta/2$ — and the two envelopes are in quadrature, so when one is at its largest the other is zero. **The energy sloshes completely from one mass to the other and back**, with envelopes exchanging roles after a time π/Δ and returning after

$$T_{\text{beat}} = \frac{2\pi}{\Delta} = \frac{2\pi}{\omega_2 - \omega_1}. \tag{0.8.56}$$

For weak coupling, expand $\omega_2 = \omega_1\sqrt{1+2\kappa/k} \approx \omega_1(1+\kappa/k)$ by Chapter 0.3, so $\Delta \approx \omega_1\kappa/k$ and $T_{\text{beat}} \approx 2\pi k/(\omega_1\kappa)$: *the weaker the coupling, the longer the beat*, diverging as $\kappa \to 0$ — which is right, since uncoupled masses never exchange anything. This is the mechanism behind two pendulums on a shared rail, a photon oscillating between two cavities, and — with the same algebra and different words — neutrino oscillation, where the two "masses" are two mass eigenstates with slightly different frequencies and the beat is measured in kilometres of flight path.



preset | mode 1 (in phase) | mode 2 (out of phase) | one mass only

coupling κ κ = 0.140 start x_1

start x_2 pause

```
ω₁ = 1.0000 ω₂ = 1.1314 Δω = 0.1314 T_beat = 47.8 s
```

```
a₁ = 0.189 a₂ = -0.494 mode energies 43.9% / 56.1%
```

Complicated motion is a superposition of simple motions. Masses $m = 1$, wall springs $k = 1$, coupling κ on the slider. *Top:* the apparatus. *Middle:* the displacements x_1 (blue) and x_2 (purple) — what you would measure. *Bottom:* the same motion in normal coordinates, $a_1 = \frac{1}{2}(x_1 + x_2)$ and $a_2 = \frac{1}{2}(x_1 - x_2)$, so that $x_1 = a_1 + a_2$ and $x_2 = a_1 - a_2$. Whatever the middle panel is doing, the bottom panel is always two pure cosines at ω_1 and ω_2 — that is (0.8.51) drawn. Press *mode 1*: only the green trace survives, and the coupling slider does nothing to it, because the coupling spring is never stretched. Press *mode 2*: only the orange trace, and now κ moves the frequency. Press *one mass only*: both modes are excited with equal amplitude, and their interference is (0.8.55) — energy sloshing from one mass to the other and back. Weaken the coupling and watch the beat period stretch out as (0.8.56) says it must, while the bottom panel barely changes. The mode energies printed on the right are $\omega_k^2 a_k(0)^2$, and they are separately constant forever: **the modes do not exchange energy, only the masses do.**

7.5 · Normal modes are principal axes

Pause on what the eigenvectors are. We observed after (0.8.50) that mM is the Hessian of V. So the normal-mode directions are the *eigenvectors of the Hessian*, which Chapter 0.6 identified as the **principal axes of curvature** of the potential surface, with eigenvalues equal to the curvatures along them. Hence

squared frequency $=$ curvature of the potential $/$ mass,

direction by direction. A stiff direction of the potential valley is a fast mode; a shallow one is a slow mode; a flat direction is a zero-frequency mode, which does not oscillate at all but drifts (you will meet one in Problem 2). "Normal modes" and "principal axes" are not two ideas that happen to use the same theorem. They are the same idea, and Chapter 6.6 will use the flat-direction version of it to make a massless particle out of a symmetry.

▼ **Grind box — N masses in a chain, solved exactly**

Nothing above used $n=2$. Take N equal masses in a line, each connected to its neighbours by springs of stiffness k, with the chain pinned to walls at both ends. Writing u_j for the displacement of mass j and setting $u_0=u_{N+1}=0$:

$$m\ddot{u}_j = k\left(u_{j+1}-2u_j+u_{j-1}\right), \qquad j=1,\ldots,N.$$

The matrix is symmetric — 2 on the diagonal, -1 next to it, zeros elsewhere — so the spectral theorem guarantees N real frequencies and N orthogonal modes before we compute anything. To find them, try $u_j=\sin(j\theta)$ for a constant θ. Then

$$u_{j+1}+u_{j-1}=\sin(j\theta+\theta)+\sin(j\theta-\theta)=2\cos\theta\ \sin j\theta,$$

by the sum-to-product identity, so the right-hand side is $k(2\cos\theta-2)u_j$ — proportional to u_j itself, which is exactly the eigenvector condition. Hence $-m\omega^2=2k(\cos\theta-1)$, and using $1-\cos\theta=2\sin^2(\theta/2)$,

$$\omega(\theta)=2\sqrt{k/m}\,\left|\sin\frac{\theta}{2}\right|.$$

The boundary condition $u_0=0$ is automatic; $u_{N+1}=0$ requires $\sin\big((N+1)\theta\big)=0$, so $\theta_n=n\pi/(N+1)$ with $n=1,\ldots,N$ (larger n repeats the same modes). Therefore

$$\omega_n=2\sqrt{\frac{k}{m}}\,\sin\left(\frac{n\pi}{2(N+1)}\right), \qquad u_j^{(n)}=\sin\left(\frac{n\pi j}{N+1}\right).$$

Check against §7.3. Put $N=2$: $\omega_1=2\sqrt{k/m}\sin(\pi/6)=\sqrt{k/m}$ and $\omega_2=2\sqrt{k/m}\sin(\pi/3)=\sqrt{3k/m}$, matching (0.8.52) at $\kappa=k$. ✓ And $u^{(1)}=(\sin\frac{\pi}{3},\sin\frac{2\pi}{3})\propto(1,1)$, $u^{(2)}=(\sin\frac{2\pi}{3},\sin\frac{4\pi}{3})\propto(1,-1)$. ✓

What to notice. N masses give exactly N modes — one per degree of freedom, which is (0.8.11) again. The mode shapes are *sampled sine waves*, with n counting half-wavelengths across the chain. And the frequencies are not evenly spaced: they crowd together near the top of the band at $2\sqrt{k/m}$, because sin flattens. That crowding is the first hint that a discrete chain and a continuous string are not the same thing at short wavelengths — the subject of §7.6.

7.6 · The limit that makes a field

Now let the chain become continuous. Give the masses a spacing a and let position along the chain be $x = ja$, so that $u_j(t)$ becomes a function $u(x,t)$ of two variables. Expand the neighbours by Taylor's theorem (Chapter 0.3):

$$u(x \pm a) = u \pm a\,\frac{\partial u}{\partial x} + \frac{a^2}{2}\frac{\partial^2 u}{\partial x^2} \pm \frac{a^3}{6}\frac{\partial^3 u}{\partial x^3} + \frac{a^4}{24}\frac{\partial^4 u}{\partial x^4} + \cdots \tag{0.8.57}$$

Add the two. Every odd term cancels — that is the payoff of using both neighbours — and

$$u(x+a) - 2u(x) + u(x-a) \;=\; a^2\,\frac{\partial^2 u}{\partial x^2} \;+\; \frac{a^4}{12}\frac{\partial^4 u}{\partial x^4} + \cdots \tag{0.8.58}$$

Substituting into the chain equation and dividing by m:

$$\frac{\partial^2 u}{\partial t^2} = \frac{ka^2}{m}\,\frac{\partial^2 u}{\partial x^2} + O(a^4). \tag{0.8.59}$$

Now take $a \to 0$ holding the *physical* quantities fixed: the mass per unit length $\rho = m/a$ and the tension $T = ka$ (the force needed for unit fractional stretch of a spring of length a). Then $ka^2/m = (ka)(a/m) = T/\rho$, a finite limit, and the correction term dies. What survives is the **wave equation**:

$$\boxed{\frac{\partial^2 u}{\partial t^2} \;=\; v^2\,\frac{\partial^2 u}{\partial x^2}, \qquad v = \sqrt{\frac{T}{\rho}}} \tag{0.8.60}$$

The discrete chain has become a continuous medium; the list of coordinates $u_1, \ldots, u_N$ has become a **field** $u(x,t)$, one number for every point of space; and the matrix M has become the operator $-v^2\partial_x^2$. Nothing else changed.

The modes survive the limit too. Look for a solution of (0.8.60) in which every point oscillates at the same frequency — that is exactly what a normal mode *is* — so put $u(x,t) = X(x)\cos\omega t$. Substituting and cancelling the time factor gives $-\omega^2 X = v^2 X''$, i.e. $X'' = -(\omega/v)^2 X$: the oscillator equation again, now in space. Its solutions are sin and cos of $\omega x/v$, and pinning the ends at $x = 0$ and $x = L$ kills the cosine and quantises the rest:

$$X_n(x) = \sin\frac{n\pi x}{L}, \qquad \omega_n = \frac{n\pi v}{L}, \qquad n = 1, 2, 3, \ldots \tag{0.8.61}$$

These are the **standing waves**, and they are the $N \to \infty$ limit of the chain's modes: setting $x = ja$ and $L = (N+1)a$ turns $u_j^{(n)} = \sin\big(n\pi j/(N+1)\big)$ into $\sin(n\pi x/L)$ exactly, while the grind box's frequency $2\sqrt{k/m}\sin\big(n\pi/2(N+1)\big)$ tends to $\sqrt{k/m}\,n\pi a/L = n\pi v/L$ for $n \ll N$. The finite chain has N modes; the field has one for every n, or equivalently one for every wavenumber $k_n = n\pi/L$. And by the argument of §7.2 — which never used the dimension — **each mode is an independent harmonic oscillator** with its own frequency ω_n, its own amplitude, its own conserved energy.

THE BRIDGE THIS WHOLE BOOK CROSSES

Assemble the chain of reasoning, because six chapters later you will need every link.

1. Any system near a stable equilibrium is quadratic in its displacements — Chapter 0.3, and Chapter 0.6's Hessian.
2. A quadratic potential gives linear equations $\ddot{\mathbf{x}} = -M\mathbf{x}$ with M symmetric — (0.8.49).
3. A symmetric matrix is diagonalisable in an orthonormal basis — Chapter 0.5's spectral theorem — so the system *is* a collection of independent harmonic oscillators, one per degree of freedom, and nothing else — (0.8.51).
4. Let the number of degrees of freedom go to infinity and the collection becomes a field, (0.8.60), whose modes are still independent oscillators, (0.8.61).

Therefore: **a field is the $N \to \infty$ limit of coupled oscillators.** That sentence is the spine of Parts V–VII. It is why quantising a field in Chapter 5.3 amounts to quantising infinitely many harmonic oscillators — you do the single-oscillator calculation of Chapter 4.5 once and then attach a copy of it to every mode. And it is why the *particle content* of quantum field theory comes out of the ladder operators of Chapter 4.5: the quantum oscillator's energy levels are evenly spaced, $E_n = (n + \frac{1}{2})\hbar\omega$, so the natural way to describe a mode's state is to say *how many quanta of $\hbar\omega$ it holds* — and those quanta, which arose as rungs on a ladder in a one-dimensional problem about a mass on a spring, are what we call particles. A photon is an excitation of one mode of the electromagnetic field; the mode is an oscillator; the photon is the integer n.

The same construction runs the rest of the book. Chapter 5.3 does it for a scalar field. Chapter 7.4 does it for a vibrating string, where the modes' quanta are the particle spectrum of string theory, and the massless spin-2 rung is the graviton. When someone tells you quantum field theory is a bewildering subject, the honest reply is that its skeleton is §7.2 with the index running to infinity.

FAMILIAR GROUND

You have read a thousand plasma concentration–time curves on semi-log axes and seen the characteristic two-slope shape: a steep early *distribution* phase, a shallower late *elimination* phase. The standard reading is that these are two processes happening one after the other. That reading is wrong in an instructive way, and §7 says exactly why.

The two-compartment model is a coupled linear system. With A_1 the amount in the central (plasma) compartment and A_2 in the peripheral tissue compartment,

$$\begin{aligned} \dot A_1 &= -\,(k_{10}+k_{12})\,A_1 + k_{21}A_2, \\ \dot A_2 &= k_{12}A_1 - k_{21}A_2, \end{aligned} \qquad \dot{\mathbf{A}} = K\,\mathbf{A}.$$

This is (0.8.49) with one time derivative instead of two, so §3 applies unchanged: seek $\mathbf{A} = \mathbf{w}\,\mathrm{e}^{\lambda t}$ and the equation becomes $K\mathbf{w} = \lambda\mathbf{w}$. **The rate constants of the biexponential curve are the eigenvalues of the rate matrix.** The characteristic polynomial is $\lambda^2 - (\operatorname{tr} K)\lambda + \det K = 0$ with $\operatorname{tr} K = -(k_{10}+k_{12}+k_{21})$ and $\det K = k_{10}k_{21}$, so writing the decay rates as $\alpha = -\lambda_1$ and $\beta = -\lambda_2$,

$$\alpha+\beta = k_{10}+k_{12}+k_{21}, \qquad \alpha\beta = k_{10}k_{21}.$$

Those are the macro-constant relations printed in every pharmacokinetics text, and they are Vieta's formulas for the roots of a quadratic. Nothing pharmacological is involved.

A concrete case. Take $k_{10} = 0.5$, $k_{12} = 0.4$, $k_{21} = 0.2$, all in h^{-1}. Then

$$K = \begin{pmatrix} -0.9 & 0.2 \\ 0.4 & -0.2 \end{pmatrix}, \qquad \lambda^2 + 1.1\lambda + 0.1 = 0,$$

$$\lambda = \frac{-1.1 \pm \sqrt{1.21-0.4}}{2} = \frac{-1.1\pm 0.9}{2} \;=\; -1.0,\; -0.1.$$

So $\alpha = 1.0\ \mathrm{h}^{-1}$ and $\beta = 0.1\ \mathrm{h}^{-1}$, with half-lives $\ln 2/\alpha = 0.69$ h and $\ln 2/\beta = 6.93$ h. The eigenvectors solve $(K-\lambda I)\mathbf{w} = 0$: for $\lambda = -1.0$, $0.1w_1 + 0.2w_2 = 0$ gives $\mathbf{w}_\alpha \propto (2,-1)$; for $\lambda = -0.1$, $-0.8w_1 + 0.2w_2 = 0$ gives $\mathbf{w}_\beta \propto (1,4)$. An intravenous bolus D into the central compartment means $\mathbf{A}(0) = (D,0)$, and expanding that in the eigenbasis, $(D,0) = \frac{4D}{9}(2,-1) + \frac{D}{9}(1,4)$, so

$$A_1(t) = \tfrac{8}{9}D\,\mathrm{e}^{-\alpha t} + \tfrac{1}{9}D\,\mathrm{e}^{-\beta t}, \qquad A_2(t) = \tfrac{4}{9}D\left(\mathrm{e}^{-\beta t} - \mathrm{e}^{-\alpha t}\right).$$

(Check: $A_1(0) = D$, $A_2(0) = 0$, and $\dot A_1(0) = -\frac{8}{9}D - \frac{0.1}{9}D = -0.9D$, which is $-(k_{10}+k_{12})D$ as the model demands. ✓)

Now the point. Both exponentials are present from $t = 0$, with fixed coefficients. Nothing switches on at any time. The curve *looks* two-phased only because one eigenvalue is ten times the other, so the fast term has essentially vanished by $t \approx 3$ h and the log-plot straightens onto the slow one. **Distribution and elimination are not two sequential biological processes. They are the two**

eigenvalues of one matrix, both operating at all times, and every fitted "phase" is a normal mode of a linear system. The terminal half-life $\ln 2/\beta$ is a property of the whole coupled system, which is exactly why it is not $\ln 2/k_{10}$ and why increasing tissue uptake k_{12} prolongs the terminal phase without changing clearance.

The one structural difference from §7, and it matters. M in (0.8.49) was symmetric because it came from a potential. K is *not* symmetric — here $K_{12} = 0.2 \neq 0.4 = K_{21}$ — because a molecule's chance of leaving plasma for tissue is not the same as its chance of coming back. So the spectral theorem does not apply, and indeed the eigenvectors are not orthogonal: $(2, -1) \cdot (1, 4) = -2 \neq 0$, an angle of $102.5°$ rather than $90°$. This is precisely the ⚠ point of Chapter 0.5's figure, where raising the skew tilted the eigen-directions away from perpendicular. The consequences are practical: you cannot read off the mode amplitudes by taking inner products as in (0.8.53) — you must solve a 2×2 system, as we did — and the modes are not energetically independent in any natural sense. Push the asymmetry far enough and the eigenvalues can collide, giving a repeated root and a $te^{-\lambda t}$ term ((0.8.18)); the fitted curve then genuinely is not a sum of two exponentials, and software that insists on fitting one will quietly report nonsense.

8 · Nonlinearity, briefly and honestly

Everything above rested on linearity. It is worth one section to see exactly what is lost when it goes, because most of the physics in the second half of this book is nonlinear.

8.1 · The pendulum, exactly

A bob of mass m on a rigid massless rod of length L, at angle θ from vertical. The restoring torque is $-mgL \sin\theta$ and the moment of inertia is mL^2, so

$$\ddot{\theta} = -\frac{g}{L}\sin\theta \;\equiv\; -\omega_0^2 \sin\theta. \tag{0.8.62}$$

This is not linear: $\sin(\theta_1 + \theta_2) \neq \sin\theta_1 + \sin\theta_2$. Chapter 0.3's move is to expand, $\sin\theta = \theta - \theta^3/6 + \cdots$, and keep the first term, giving $\ddot{\theta} = -\omega_0^2\theta$ and $T_0 = 2\pi\sqrt{L/g}$. The question worth asking is not *whether* that is an approximation — obviously it is — but *how good*, and where it stops being good at all.

The energy first integral of §4.3 still works, because multiplying by $\dot{\theta}$ does not care about linearity: multiply (0.8.62) by $\dot{\theta}$ and integrate,

$$\tfrac{1}{2}\dot{\theta}^2 - \omega_0^2\cos\theta \;=\; \mathcal{E} \;=\; \text{constant}. \tag{0.8.63}$$

Releasing from rest at θ_0 fixes $\mathcal{E} = -\omega_0^2\cos\theta_0$, and the grind box turns this into an exact period. The answer is $T = T_0 \cdot \frac{2}{\pi} K\left(\sin\frac{\theta_0}{2}\right)$ with K the complete elliptic integral, and expanding it,

$$\frac{T}{T_0} = 1 + \frac{\theta_0^2}{16} + \frac{11\,\theta_0^4}{3072} + \cdots \tag{0.8.64}$$

The period depends on amplitude. That single fact is the death of superposition: two solutions of different sizes have different frequencies, so their sum is not a solution of anything. Numerically, the correction is $+0.19\%$ at $\theta_0 = 10^\circ$, $+1.74\%$ at 30°, and $+18.0\%$ at 90° — which is why a pendulum clock has a small amplitude and why Huygens had to think hard about escapements.

8.2 · The phase portrait and the separatrix

Plot the state $(\theta, \dot{\theta})$, as in §4.4. From (0.8.63) the trajectories are the level curves of $\mathcal{E}$, and there are two qualitatively different families. For $\mathcal{E} < \omega_0^2$ the curve closes: the bob swings back and forth, and near the origin the closed curve is the ellipse of §4.4, slightly distorted. For $\mathcal{E} > \omega_0^2$ there is enough energy to pass over the top, $\dot{\theta}$ never vanishes, and the trajectory runs off to $\theta \to \infty$ — the pendulum whirls.

Between them sits the single curve $\mathcal{E} = \omega_0^2$, on which the bob arrives at the inverted position with exactly zero speed. Substituting into (0.8.63) and using $1 + \cos\theta = 2\cos^2(\theta/2)$:

$$\dot{\theta} = \pm 2\omega_0 \cos\frac{\theta}{2}. \tag{0.8.65}$$

This is the **separatrix**, and it does something the linear oscillator cannot: it takes infinite time to complete. Near $\theta = \pi$ the right-hand side vanishes linearly in $(\pi - \theta)$, so the approach is exponential and the destination is never reached. The inverted position is an unstable equilibrium — a maximum of the potential, a negative Hessian eigenvalue in Chapter 0.6's classification, and hence an $\mathrm{e}^{+\omega_0 t}$ in §3's language rather than an oscillation. Linearising about the *top* of the potential gives you exponential growth, which is the correct local answer; there is nothing wrong with linearisation except that it is local.

8.3 · Chaos, in one paragraph

Add a drive and damping to (0.8.62) and it becomes possible for trajectories to be bounded, aperiodic, and exponentially sensitive to initial conditions: two starting points a distance ϵ apart separate like $\epsilon\,\mathrm{e}^{\Lambda t}$ for some positive Λ. Picard–Lindelöf is untouched — the solution still exists and is still unique — but predicting the state at time t to fixed accuracy now requires knowing the initial state to accuracy $\mathrm{e}^{-\Lambda t}$, so every factor of e in your instruments buys one more unit of $1/\Lambda$ in forecast horizon. That is the entire content of "chaos": determinism and predictability are different properties, and only the first is guaranteed. It cannot happen in a linear system, where solutions are sums of $\mathrm{e}^{\lambda_i t}$ and neighbouring trajectories either converge, or diverge in a fixed direction, or wind around each other forever, but never explore.

⚠ WHY THIS ISN'T OBVIOUS

Superposition is a privilege, not a law of nature. Every technique in this chapter that felt like the natural way to think — decompose into modes, solve each separately, add the answers; expand a source into sinusoids and treat them one at a time; write the general solution as a basis with arbitrary coefficients — rests on (0.8.9), one line of algebra that is true only because L is linear. The habit is so ingrained that it is easy to forget it can fail. It fails constantly.

General relativity is nonlinear (Chapter 3.6). The Einstein equations relate curvature to energy, and gravitational fields carry energy, so *gravity gravitates*: the field is a source for itself. You may not add the field of two stars to get the field of the pair. The equations are not merely algebraically messier — the superposition principle is simply unavailable, which is why the two-body problem in general relativity required supercomputers and a half-century of work while in Newtonian gravity it is an exercise.

Yang–Mills theory is nonlinear (Chapter 6.4). A photon carries no electric charge, so electromagnetism is linear and light beams pass through each other. A gluon *does* carry colour charge, so gluons interact with gluons directly. That one difference is the reason the strong force confines and the electromagnetic force does not, and the reason nobody can write down the spectrum of QCD in closed form.

What replaces superposition when it is gone? Almost always **perturbation theory**: find a linear problem nearby, solve it exactly, and treat the nonlinearity as a small correction computed order by order. Chapter 5.6 builds that machinery, and Feynman diagrams are its bookkeeping — each vertex in a diagram is one factor of the nonlinear term. So the harmonic oscillator is not merely the first system we can solve; it is the system every hard problem is expanded around. That is why this chapter is long, and it is why the small-oscillations result of Chapter 0.3 is arguably the most-used theorem in physics.

▼ **Grind box — the exact pendulum period**

From (0.8.63) with $\mathcal{E} = -\omega_0^2 \cos\theta_0$,

$$\dot{\theta} = \omega_0 \sqrt{2\left(\cos\theta - \cos\theta_0\right)}.$$

A quarter period is the time from $\theta = 0$ to $\theta = \theta_0$, so separating variables,

$$\frac{T}{4} = \int_0^{\theta_0} \frac{\mathrm{d}\theta}{\omega_0 \sqrt{2(\cos\theta - \cos\theta_0)}}.$$

Use $\cos\theta = 1 - 2\sin^2(\theta/2)$ to write $\cos\theta - \cos\theta_0 = 2\big(s^2 - \sin^2(\theta/2)\big)$ where $s \equiv \sin(\theta_0/2)$. Then substitute $\sin(\theta/2) = s\sin\varphi$, which maps $\theta : 0 \to \theta_0$ onto $\varphi : 0 \to \pi/2$ and gives $\frac{1}{2}\cos(\theta/2)\,\mathrm{d}\theta = s\cos\varphi\,\mathrm{d}\varphi$ with $\cos(\theta/2) = \sqrt{1 - s^2\sin^2\varphi}$. The square root in the denominator becomes $2s\cos\varphi$, and it cancels against the numerator exactly:

$$\frac{T}{4} = \frac{1}{\omega_0} \int_0^{\pi/2} \frac{\mathrm{d}\varphi}{\sqrt{1 - s^2 \sin^2 \varphi}} \equiv \frac{K(s)}{\omega_0}.$$

K is the complete elliptic integral of the first kind — a function defined by this integral, which cannot be reduced to elementary ones. Since $T_0 = 2\pi/\omega_0$, $T/T_0 = 2K(s)/\pi$, and $K(0) = \pi/2$ recovers $T = T_0$. ✓

The expansion. Expand the integrand by the binomial series (Chapter 0.3), $(1-u)^{-1/2} = 1 + \frac{1}{2}u + \frac{3}{8}u^2 + \cdots$ with $u = s^2 \sin^2 \varphi$, and integrate term by term using $\int_0^{\pi/2} \sin^2 \varphi \, \mathrm{d}\varphi = \pi/4$ and $\int_0^{\pi/2} \sin^4 \varphi \, \mathrm{d}\varphi = 3\pi/16$:

$$K(s) = \frac{\pi}{2} \left(1 + \frac{s^2}{4} + \frac{9s^4}{64} + \cdots \right).$$

Finally put $s = \sin(\theta_0/2) = \frac{\theta_0}{2} - \frac{\theta_0^3}{48} + \cdots$, so $s^2 = \frac{\theta_0^2}{4} - \frac{\theta_0^4}{48} + \cdots$ and $s^4 = \frac{\theta_0^4}{16} + \cdots$, and collect:

$$\frac{T}{T_0} = 1 + \frac{\theta_0^2}{16} + \left(\frac{9}{64 \cdot 16} - \frac{1}{4 \cdot 48} \right) \theta_0^4 + \cdots = 1 + \frac{\theta_0^2}{16} + \frac{11\theta_0^4}{3072} + \cdots$$

since $\frac{9}{1024} - \frac{1}{192} = \frac{27-16}{3072} = \frac{11}{3072}$, confirming (0.8.64). At $\theta_0 = 1$ rad the series gives 1.06608 against the exact 1.06633 — the next term is doing 2×10^{-4} of work, as an $O(\theta_0^6)$ term should.

9 · Worked examples

WORKED EXAMPLE 1 — TWO MASSES, FROM MATRIX TO BEATS

Two masses $m = 1\,\mathrm{kg}$, *wall springs* $k = 1\,\mathrm{N/m}$, *coupling* $\kappa = 0.1\,\mathrm{N/m}$. *Mass 1 is displaced* $10\,\mathrm{cm}$ *and both are released from rest. Find everything.*

The matrix. From (0.8.49),

$$M = \begin{pmatrix} 1.1 & -0.1 \\ -0.1 & 1.1 \end{pmatrix}\ \mathrm{s}^{-2}.$$

Eigenvalues. $\det(M - \lambda I) = (1.1 - \lambda)^2 - 0.01 = 0$ gives $1.1 - \lambda = \pm 0.1$, so $\lambda = 1.0$ and $1.2\ \mathrm{s}^{-2}$ — real, as the spectral theorem guarantees for a symmetric matrix. Hence

$$\omega_1 = 1.0000\ \mathrm{s}^{-1}, \qquad \omega_2 = \sqrt{1.2} = 1.0954\ \mathrm{s}^{-1}.$$

Check against (0.8.52): $\omega_1^2 = k/m = 1$ ✓ and $\omega_2^2 = (k + 2\kappa)/m = 1.2$ ✓.

Eigenvectors. For $\lambda = 1.0$: $(M - I)\mathbf{w} = \begin{pmatrix} 0.1 & -0.1 \\ -0.1 & 0.1 \end{pmatrix} \mathbf{w} = 0$ gives $w_1 = w_2$, so $\mathbf{u}_1 = \frac{1}{\sqrt{2}}(1, 1)$. For $\lambda = 1.2$: $-0.1 w_1 - 0.1 w_2 = 0$ gives $w_1 = -w_2$, so $\mathbf{u}_2 = \frac{1}{\sqrt{2}}(1, -1)$. Their dot product is $\frac{1}{2}(1 - 1) = 0$: orthogonal, unbidden.

General solution. By (0.8.53),

$$\mathbf{x}(t) = \left(a_1 \cos\omega_1 t + \tfrac{b_1}{\omega_1} \sin\omega_1 t\right) \mathbf{u}_1 + \left(a_2 \cos\omega_2 t + \tfrac{b_2}{\omega_2} \sin\omega_2 t\right) \mathbf{u}_2.$$

Initial conditions. $\mathbf{x}(0) = (0.1, 0)\ \mathrm{m}$, $\dot{\mathbf{x}}(0) = \mathbf{0}$. So $b_k = 0$ and

$$a_1 = \mathbf{u}_1 \cdot \mathbf{x}(0) = \frac{0.1}{\sqrt{2}}, \qquad a_2 = \mathbf{u}_2 \cdot \mathbf{x}(0) = \frac{0.1}{\sqrt{2}}.$$

Equal amplitudes in both modes — which is the content of "displace one mass": in the mode basis, a single-mass displacement is an equal superposition. Assembling components,

$$x_1(t) = 0.05\,(\cos\omega_1 t + \cos\omega_2 t)\,, \qquad x_2(t) = 0.05\,(\cos\omega_1 t - \cos\omega_2 t)\ \ \mathrm{m}.$$

The beat. With $\Delta = \omega_2 - \omega_1 = 0.09544\ \mathrm{s}^{-1}$ and $\bar{\omega} = 1.04772\ \mathrm{s}^{-1}$, (0.8.55) gives

$$\begin{aligned} x_1 &= 0.1\ \cos(0.04772\,t)\ \cos(1.04772\,t), \\ x_2 &= 0.1\ \sin(0.04772\,t)\ \sin(1.04772\,t). \end{aligned}$$

Mass 1 is momentarily still and mass 2 has all the motion at $t = \pi/\Delta = 32.9$ s, and the pattern repeats with

$$T_{\text{beat}} = \frac{2\pi}{\Delta} = 65.8\ \mathrm{s},$$

against the individual oscillation period $2\pi/\bar{\omega} = 6.00$ s — eleven fast cycles per exchange. The weak-coupling estimate $2\pi k/(\omega_1 \kappa) = 62.8$ s is 5% low, which is the size of the next term in $\sqrt{1 + 2\kappa/k}$, as it should be.

Energy accounting. Total energy is $\frac{1}{2}kx_1^2 + \frac{1}{2}kx_2^2 + \frac{1}{2}\kappa(x_1 - x_2)^2$ evaluated at $t = 0$, which is $\frac{1}{2}(1)(0.01) + \frac{1}{2}(0.1)(0.01) = 5.50$ mJ. In the mode picture it is $\frac{1}{2}\omega_1^2 q_1(0)^2 + \frac{1}{2}\omega_2^2 q_2(0)^2$ with $q_k(0) = 0.1/\sqrt{2}$, giving $\frac{1}{2}(1.0 + 1.2)(0.005) = 5.50$ mJ ✓ — split 45.5% / 54.5% between the modes, and those two numbers never change, no matter how violently the energy moves between the two *masses*.

WORKED EXAMPLE 2 — A DRIVEN DAMPED OSCILLATOR, PEAK AND WIDTH

Take $\omega_0 = 1\ \mathrm{s}^{-1}$ and $\gamma = 0.05\ \mathrm{s}^{-1}$, so $Q = 10$. Locate the amplitude peak, the power peak, and the width; compare the Lorentzian.

Amplitude. From (0.8.40),

$$\omega_{\text{peak}} = \sqrt{1 - 2(0.05)^2} = \sqrt{0.995} = 0.99750\ \mathrm{s}^{-1},$$

which is *below* ω_0 by 0.25% — and below the damped free frequency $\omega_{\mathrm{d}} = \sqrt{1 - 0.0025} = 0.99875$ as well. Three different "resonant frequencies", all distinct, all correct answers to different questions. The peak amplitude is

$$A_{\max} = \frac{F_0}{2(0.05)\sqrt{1 - 0.0025}} = 10.0125\, F_0,$$

versus the static response $A(0) = F_0/\omega_0^2 = F_0$. The system amplifies by very nearly Q, which is (0.8.40) in the limit $\gamma \ll \omega_0$: $A_{\max} \to F_0/(2\gamma\omega_0) = Q\, A(0)$.

Power. By (0.8.43) the power peak is at $\omega = \omega_0 = 1$ exactly, with $\bar{P} = F_0^2/4\gamma = 5F_0^2$. The half-power points from (0.8.44):

$$\omega_\pm = \sqrt{1 + 0.0025} \pm 0.05 = 1.00125 \pm 0.05 = 1.05125,\ 0.95125,$$

so $\mathrm{FWHM} = 0.10000 = 2\gamma$ exactly, and $\omega_+\omega_- = 1.05125 \times 0.95125 = 1.00000 = \omega_0^2$ ✓. Note that the centre of the half-power interval is 1.00125, not $\omega_0 = 1$ — the curve is very slightly asymmetric, and the correct recipe for extracting ω_0 from data is the geometric mean, not the arithmetic one.

The Lorentzian. Near resonance (0.8.47) reads

$$\frac{\bar{P}(\omega)}{\bar{P}(\omega_0)} \approx \frac{0.0025}{(\omega - 1)^2 + 0.0025}.$$

Compare it with the exact (0.8.42) at three places. At $\omega = 1.05$, just inside the half-power point, the Lorentzian gives 0.5000 against the exact 0.5120 — 2.4% low. At $\omega = 1.5$ it gives 0.0099 against 0.0142, now 30% low. And at $\omega = 0.5$ it gives 0.0099 again — because a Lorentzian is exactly symmetric about ω_0 — while the true answer is 0.0044, less than half. **The real resonance is asymmetric and the Lorentzian is not**, which is another reading of $\omega_+\omega_- = \omega_0^2$: the true curve is symmetric in $\log\omega$, not in ω. So the Lorentzian is a statement about the neighbourhood of the peak and nothing else — which is exactly the regime in which particle physics uses it, and exactly why a Breit–Wigner fit is quoted with a fit window.

Width and lifetime. Free ringdown has energy $\mathrm{e}^{-2\gamma t}$, an energy lifetime $\tau = 1/2\gamma = 10$ s. Multiply by the width: $\mathrm{FWHM} \times \tau = 0.1 \times 10 = 1$. Reading that with $E = \hbar\omega$ turns it into $\Gamma\tau = \hbar$. Everything in this example is a laboratory-bench version of how a resonance mass and width are quoted in a particle data table.

10 · Your turn

PROBLEM 1 — WHY CRITICAL DAMPING IS FORCED TO GROW A T

Solve $\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = 0$ in the critically damped case $\gamma = \omega_0$. Show that $\mathrm{e}^{-\gamma t}$ alone cannot be the general solution, produce the second solution without guessing it, and verify it satisfies the equation.

− Solution

Why one solution is not enough. The equation is second order, so by (0.8.11) its solution space is two-dimensional. The multiples of $\mathrm{e}^{-\gamma t}$ form a one-dimensional subspace. Something is missing, and it is not optional: with only $x = C\mathrm{e}^{-\gamma t}$ available you could not satisfy the perfectly ordinary initial conditions $x(0) = 0, \dot{x}(0) = v_0 \neq 0$, since $x(0) = 0$ forces $C = 0$.

Producing it. The characteristic equation is $\lambda^2 + 2\gamma\lambda + \gamma^2 = (\lambda + \gamma)^2 = 0$, a double root at $-\gamma$. In operator form, $(D + \gamma)^2 x = 0$. Set $N = D + \gamma$. We need a function with $Nw \neq 0$ but $N^2 w = 0$, i.e. one that N maps *onto* the eigenfunction rather than to zero. From (0.8.18),

$$N\left(t\mathrm{e}^{-\gamma t}\right) = \frac{\mathrm{d}}{\mathrm{d}t}\left(t\mathrm{e}^{-\gamma t}\right) + \gamma t\mathrm{e}^{-\gamma t} = \mathrm{e}^{-\gamma t} - \gamma t\mathrm{e}^{-\gamma t} + \gamma t\mathrm{e}^{-\gamma t} = \mathrm{e}^{-\gamma t},$$

and applying N again gives $N\mathrm{e}^{-\gamma t} = 0$. So $t\mathrm{e}^{-\gamma t}$ is annihilated by N^2 and is the second solution. Independence is clear since their ratio t is not constant.

Direct check. With $x = t\mathrm{e}^{-\gamma t}$: $\dot{x} = (1 - \gamma t)\mathrm{e}^{-\gamma t}$ and $\ddot{x} = (-2\gamma + \gamma^2 t)\mathrm{e}^{-\gamma t}$. Then

$$\ddot{x} + 2\gamma\dot{x} + \gamma^2 x = \left[-2\gamma + \gamma^2 t + 2\gamma - 2\gamma^2 t + \gamma^2 t\right]\mathrm{e}^{-\gamma t} = 0. \checkmark$$

Why it had to happen. Two routes, both in §3.4. Algebraically, N restricted to the solution space satisfies $N^2 = 0 \neq N$, so it is nilpotent and cannot be diagonalised; its matrix is the Jordan block (0.8.19) and $t\mathrm{e}^{-\gamma t}$ is the generalised eigenvector. Analytically, take the underdamped solution and let $\omega_\mathrm{d} \to 0$: $\sin(\omega_\mathrm{d} t)/\omega_\mathrm{d} \to t$, so the second basis vector does not vanish, it degenerates into $t\mathrm{e}^{-\gamma t}$. The factor of t is the fingerprint of two eigenvalues colliding, and it appears for the same reason at exact resonance in §6.

PROBLEM 2 — THREE FREE MASSES, AND A MODE THAT DOES NOT OSCILLATE

Three equal masses m lie in a line, joined by two springs of stiffness k, with *nothing* attached to any wall. Write the equations of motion, find the three normal modes and frequencies, and interpret the mode with $\omega = 0$. What conservation law is it?

− Solution

Equations. Only the two springs act, each pulling its two neighbours together:

$$m\ddot{x}_1 = k(x_2 - x_1), \quad m\ddot{x}_2 = k(x_1 - x_2) + k(x_3 - x_2), \quad m\ddot{x}_3 = k(x_2 - x_3),$$

$$\ddot{\mathbf{x}} = -\frac{k}{m}C\mathbf{x}, \qquad C = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}.$$

C is symmetric, so §7.2 applies: three real eigenvalues, three orthogonal modes.

Eigenvalues. Guess the symmetric candidates and verify, which is faster than expanding a cubic. $C(1,1,1)^\mathsf{T} = (0,0,0)^\mathsf{T}$, so $\mu = 0$. $C(1,0,-1)^\mathsf{T} = (1,0,-1)^\mathsf{T}$, so $\mu = 1$. $C(1,-2,1)^\mathsf{T} = (3,-6,3)^\mathsf{T} = 3(1,-2,1)^\mathsf{T}$, so $\mu = 3$. Three eigenvalues of a 3×3 matrix, and their sum $0 + 1 + 3 = 4$ matches $\operatorname{tr} C = 1 + 2 + 1$ ✓. The three vectors are mutually orthogonal, as promised. Hence

$$\omega_1 = 0, \qquad \omega_2 = \sqrt{k/m}, \qquad \omega_3 = \sqrt{3k/m}.$$

Shapes. Mode 2, $(1,0,-1)$: the ends move oppositely and the centre stays still, so it is a node — the centre mass feels equal and opposite pulls. Mode 3, $(1,-2,1)$: the ends move together against the centre, stretching both springs, hence the highest frequency; note the centre moves twice as far, which is exactly what keeps the centre of mass fixed ($1 - 2 + 1 = 0$).

The zero mode. $\omega = 0$ means $\ddot{q}_1 = 0$, so $q_1 = a + bt$: not an oscillation but a *drift*. The mode shape $(1,1,1)$ is uniform translation — sliding the whole chain sideways. It costs no potential energy because no spring changes length, so the potential is perfectly flat in that direction, and a flat direction has zero Hessian eigenvalue (Chapter 0.6's fourth case, where the second-derivative test goes silent).

The conservation law. $q_1 \propto x_1 + x_2 + x_3$ is the centre-of-mass coordinate, so $\dot{q}_1 =$ constant says the **total momentum is conserved**. And the reason the potential is flat along $(1,1,1)$ is that V depends only on the differences $x_{j+1} - x_j$ — that is, V is unchanged by translating the whole system. So: *a symmetry of the potential (translation invariance) produced a flat direction, which produced a zero-frequency mode, which is a conservation law (momentum).* That chain is **Noether's theorem**, and Chapter 1.4 proves it in general. Here you have met it in a 3×3 matrix. Zero modes from symmetry recur throughout the book, most consequentially in Chapter 6.6, where the flat directions of the Higgs potential are massless particles and the entire drama is what happens to them.

PROBLEM 3 — A BIEXPONENTIAL CURVE IS AN EIGENVALUE PROBLEM

A two-compartment model has $k_{10} = 1.0, k_{12} = 0.75, k_{21} = 0.5$, all in h^{-1}. Find α and β and the two half-lives; find the eigenvectors and check they are not orthogonal; and write $A_1(t)$ after an intravenous bolus D.

− Solution

The matrix and its spectrum.

$$K = \begin{pmatrix} -(k_{10}+k_{12}) & k_{21} \\ k_{12} & -k_{21} \end{pmatrix} = \begin{pmatrix} -1.75 & 0.5 \\ 0.75 & -0.5 \end{pmatrix}.$$

$\operatorname{tr} K = -2.25$ and $\det K = 0.875 - 0.375 = 0.5$ — which is $k_{10}k_{21}$, as the general identity says. So $\lambda^2 + 2.25\lambda + 0.5 = 0$ and

$$\lambda = \frac{-2.25 \pm \sqrt{5.0625 - 2}}{2} = \frac{-2.25 \pm 1.75}{2} = -2.00,\ -0.25\ \mathrm{h}^{-1}.$$

Hence $\alpha = 2.00\ \mathrm{h}^{-1}$, $\beta = 0.25\ \mathrm{h}^{-1}$. Sanity check on the macro-constant relations: $\alpha + \beta = 2.25 = k_{10} + k_{12} + k_{21}$ ✓ and $\alpha\beta = 0.5 = k_{10}k_{21}$ ✓. Half-lives:

$$t^{\alpha}_{1/2} = \frac{\ln 2}{2.00} = 0.347\ \mathrm{h} = 20.8\ \mathrm{min}, \qquad t^{\beta}_{1/2} = \frac{\ln 2}{0.25} = 2.77\ \mathrm{h}.$$

Note $t^{\beta}_{1/2}$ is not $\ln 2/k_{10} = 0.69$ h. The terminal half-life is a property of the coupled pair, four times longer than elimination alone would suggest, because drug that went into tissue has to come back before it can leave.

Eigenvectors. For $\lambda = -2$: $(K + 2I)\mathbf{w} = \begin{pmatrix} 0.25 & 0.5 \\ 0.75 & 1.5 \end{pmatrix}\mathbf{w} = 0$ gives $w_1 = -2w_2$, so $\mathbf{w}_\alpha \propto (2, -1)$. For $\lambda = -0.25$: $\begin{pmatrix} -1.5 & 0.5 \\ 0.75 & -0.25 \end{pmatrix}\mathbf{w} = 0$ gives $w_2 = 3w_1$, so $\mathbf{w}_\beta \propto (1, 3)$. Their dot product is $2 - 3 = -1 \neq 0$: **not orthogonal** (the angle is $98.1°$), because K is not symmetric — $k_{21} = 0.5$ while $k_{12} = 0.75$. Chapter 0.5's spectral theorem simply does not apply, so we must solve for the coefficients rather than take inner products.

Bolus. $\mathbf{A}(0) = (D, 0)$. Write $(D, 0) = a(2, -1) + b(1, 3)$: from the second component $a = 3b$, and the first gives $6b + b = D$, so $b = D/7$ and $a = 3D/7$. Therefore

$$A_1(t) = \tfrac{6}{7}D\,\mathrm{e}^{-2t} + \tfrac{1}{7}D\,\mathrm{e}^{-0.25t}, \qquad A_2(t) = \tfrac{3}{7}D\left(\mathrm{e}^{-0.25t} - \mathrm{e}^{-2t}\right).$$

Checks: $A_1(0) = D$ ✓, $A_2(0) = 0$ ✓, and $\dot{A}_1(0) = -\frac{12}{7}D - \frac{0.25}{7}D = -1.75D = -(k_{10} + k_{12})D$ ✓.

Reading it. 86% of the initial plasma amount decays with the fast eigenvalue and 14% with the slow one, both starting at $t = 0$. Plot $\ln A_1$ and you see a steep segment bending onto a shallow line whose back-extrapolated intercept is $D/7$ — which is how the classical "method of residuals" recovers the coefficients by hand. It is peeling eigenvectors off one at a time, starting with the one that survives longest.

PROBLEM 4 — ANYTHING CAN BE A WAVE, AND WHAT THAT SAYS ABOUT CAUSALITY

Show that $u(x,t) = f(x - vt)$ solves the wave equation (0.8.60) for *any* twice-differentiable f. Show the same for $g(x + vt)$, and argue that the general solution is the sum. Then say what this implies about how information travels.

– Solution

Direct substitution. Write $s = x - vt$. By the chain rule (Chapter 0.6), $\partial u/\partial x = f'(s)$ and $\partial^2 u/\partial x^2 = f''(s)$, while $\partial u/\partial t = -vf'(s)$ and $\partial^2 u/\partial t^2 = v^2 f''(s)$. Hence

$$\frac{\partial^2 u}{\partial t^2} - v^2 \frac{\partial^2 u}{\partial x^2} = v^2 f''(s) - v^2 f''(s) = 0 \checkmark$$

for every f. The sign of v never mattered, so $g(x + vt)$ works identically, and sums work because the equation is linear.

Why that is everything. Change variables to $\xi = x - vt, \eta = x + vt$. Then $\partial_x = \partial_\xi + \partial_\eta$ and $\partial_t = v(\partial_\eta - \partial_\xi)$, so

$$\frac{\partial^2}{\partial t^2} - v^2 \frac{\partial^2}{\partial x^2} = v^2 \left(\partial_\eta - \partial_\xi\right)^2 - v^2 \left(\partial_\xi + \partial_\eta\right)^2 = -4v^2\, \partial_\xi \partial_\eta,$$

because the squared terms cancel and the cross terms add. So the wave equation is simply $\partial_\xi \partial_\eta u = 0$, whose solutions are exactly $u = f(\xi) + g(\eta)$: integrate in η to get $\partial_\xi u =$ (function of ξ alone), then integrate in ξ. This is **d'Alembert's solution**, and the count is right — two arbitrary functions for a second-order equation, the field-theoretic version of two arbitrary constants.

What it means. The shape f is completely arbitrary and it moves rigidly to the right at speed v, undistorted. So a disturbance made here and now is felt *there* exactly at the moment $|x - x_0|/v$ later — not sooner, and with no precursor. The wave equation carries information at exactly one speed, and that speed is a property of the medium ($v = \sqrt{T/\rho}$), not of the source or the observer.

That last clause is the entire crisis of 1900. Chapter 2.1 will derive a wave equation for the electromagnetic field straight out of Maxwell's equations, with $v = 1/\sqrt{\varepsilon_0 \mu_0} = c$ — a speed built from two laboratory constants and referring to no medium anybody could find. Combined with the argument above, that forces the conclusion that c is the same for every observer, which is Einstein's second postulate. And the structure $f(x - vt)$ is why Chapter 2.3 draws light cones: the set of events reachable from here is exactly the set the solutions of (0.8.60) can reach. Causality, in this book, is a statement about which arguments a function is allowed to depend on.

THE BRICK YOU JUST LAID

You can now solve every linear differential equation with constant coefficients, and you know why the method works: $\mathrm{e}^{\lambda t}$ **is the eigenfunction of** $\mathrm{d}/\mathrm{d}t$, so a linear ODE is an eigenvalue problem and its solution set is a vector space whose dimension is the order. Repeated roots are defective matrices; $t\mathrm{e}^{\lambda t}$ is a generalised eigenvector and the Jordan block of Chapter 0.5 is where it lives. You solved the harmonic oscillator three ways, read its phase-space ellipse as energy conservation, and computed the area that Chapter 4.5 will quantise. You classified damping by a discriminant, defined Q twice — radians per e-fold of energy, and centre frequency over width — and proved those two definitions are the same number. You derived the resonance curve, showed the power peak is exactly at ω_0 with width exactly 2γ, and recognised the Lorentzian. And you diagonalised a coupled system into normal modes, watched a general motion resolve into two simple ones, and took $N \to \infty$ to get a field.

Where this gets spent.

- **Eigenfunctions of** $\mathrm{d}/\mathrm{d}t$ → Chapter 0.9, where diagonalising the derivative on the whole line *is* the Fourier transform; Chapter 4.5, where energy eigenstates are the same construction for $\hat{H}$; Chapter 5.3, where field modes are the same construction again.
- **The harmonic oscillator** → Chapter 4.5 (ladder operators, and the $\frac{1}{2}\hbar\omega$ that will not go away), Chapter 5.3 (one oscillator per field mode), Chapter 7.4 (one oscillator per string mode, whose quanta are the particle spectrum).
- **The Lorentzian and** Q → Chapter 5.7, where the same curve is the Breit–Wigner formula and the width of a bump in a cross-section *is* the inverse lifetime of a particle nobody could ever watch decay.
- **Normal modes** → Chapter 1.3 (small oscillations in phase space), Chapter 5.3 and Chapter 7.4 (infinitely many of them), Chapter 6.6 (a zero-frequency mode is a massless particle).
- **The wave equation** → Chapter 2.1, where Maxwell's equations produce it with $v = c$ and special relativity becomes unavoidable; Chapter 2.3 (light cones); Chapter 4.4, where adding an i turns it into the Schrödinger equation.
- **Duhamel and the Green's function** → Chapter 0.9 (convolution and δ), Chapter 5.6 (propagators, which are this integral with four-vectors in it).
- **Nonlinearity** → Chapter 3.6 (gravity gravitates), Chapter 6.4 (gluons carry colour), Chapter 5.6 (perturbation theory, the only general tool once superposition is gone).

One sentence to carry above all others: *a field is infinitely many coupled oscillators, and the spectral theorem is what lets you treat them one at a time.* Everything from Chapter 5.3 onward is that sentence being cashed. Chapter 0.9 supplies the last piece — how to expand an arbitrary function in modes when there is a continuum of them — and Part 0 is finished.